Stock exchange smart guide

Stock Exchange Smart Guide

1. Introduction

1.1. The stock exchange

Stock exchanges are public, centralised, and organised markets. The Hungarian Stock Exchange is a legal entity with a sovereign council. Although the institute is not profit oriented, it finances its operations from the revenues received. The goods or shares circulating on the stock exchange do not need to be present physically, as the items are replaceable and interchangeable with each other. There are two types of exchange markets: commodity (or mercantile) and stock exchange. Within the stock exchange, share and currency exchanges can be distinguished.

Therefore, stock exchange is the market of replaceable bulk goods, where trade is organised in accordance with specific practices. **Based on legal status, there are two types of stock exchanges:**

- The European-Continental stock exchanges are monitored by the government and are based on Public Law.
- The Anglo-Saxon type stock exchanges are based on Private Law. The institutes usually operate as public companies.

There are 235 stock exchanges worldwide, operating in 128 countries. Regionally the number of stock exchanges are the following: 71 in Europe, 83 in Asia, 33 in North America, 16 in South America, 25 in Africa, and 7 in Oceania.

1.2. Securities

The legal term of the securities is defined by the Civil Code (called Polgári Törvénykönyv (Ptk.) in Hungary). A security is a financial instrument or a collection of data recorded, and transmitted (in compliance with the definition in the Civil Code). It must meet the determined prerequisites, and have a legally accepted form. (Ptk. 338/A §)

Securities - from a financial aspect - are promises to pay on a future date. In this sense (but not legally), derivatives like forwards& futures, options, and swaps, are securities.

Securities have two forms: **documents and dematerialised securities.** The latter consists of a large set of data, containing all legal criteria of a security. It is created, recorded, and transmitted electronically, as defined by law and other regulations.

Securities can be categorised into three big groups. The first category of securities **embodies a claim (debt securities).** Typically, none of the parties in the contract know their obligations towards the other party exactly, yet agree to pay all their debts within a predefined period. The most common types of such claims are bills of exchange, checks, bonds, and deposit certificates.

The next category of securities **deals with rights towards assets.** They grant the right of disposal to the owner of the security. Warehouse warrants are the most common example in this group. The last category includes **securities granting co-ownership rights (equity securities)**. They certify that the owner of the security has contributed to the capital resources of a given firm. Thus, entitle the owner to receive dividends in proportion of his or her ownership in the firm. Owners are usually forbidden to withdraw their funds from the business. The most common security in this group is the share of companies, but investment funds also belong here.

1.3. Banks, brokers, brokerage firms

Banks are the most important participants of the financial intermediary system. By definition, a bank is an institute offering financial services. Its two main activities are: providing loans and collecting deposits. Banks collect the savings currently not in use (passive banking) and pump it into the part of the economy where funds are needed (active banking). Thus, banking is the motor of the modern, capital-based economy. Banks have two main revenue sources: fees of the different financial services and interest rates of loans.

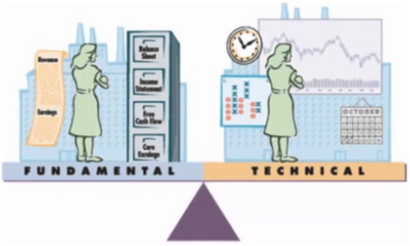
A broker is someone who buys and sells goods or assets on behalf of his or her customers in exchange of a brokerage fee. The deals can be executed both on the stock exchange and outside the stock exchange. Brokers also trade with securities and may be entitled to execute different account transactions. According to the new Hungarian securities regulations, the former requires a registered capital of HUF 20 million, while the latter requires HUF 100 million. Besides the required capital, a professional (the broker) must be nominated to act on behalf of the investor. In order to execute the orders of the clients, the broker must have a professional securities examination certificate, have at least two years of professional experience on the securities market, and must be morally irreproachable. In Hungary, the activity of trading with securities must be executed through specialised companies. However, on the commodities exchange and on the futures market of the Budapest Stock Exchange, individual brokers are also allowed to become members.

The main task of brokerage firms is to act as an agent for securities trading. This means that the investment service provider acts in its on name, but on behalf of another person (the customer) to sell or buy. The investor enters into a contract of agency with the chosen brokerage for every single sell or buy order. The contract of agency must clearly specify the financial product to be traded, its price and quantity, and the nature of the order (sell or buy). The legal and financial conditions of the orders are recorded in the general business rules of the brokerage (complying with statutory requirements). The fee of the services may vary for the different brokerage firms.

The two main types of orders made by the investor are the following:

- In case of a sell order, the customer sells his or her securities. If the security has a physical form, it needs to be handed over. If the security exists on an account, it should be transferred to the brokerage firm's account.
- In case of a buy order, the customer buys securities and must pay in exchange accordingly. The amount to be paid consists of two parts: the price of the security and the commission of the agency. In both cases the settlement happens only once the order has been fulfilled.

1.4. Fundamental and technical analysis



Market processes need to be examined in order to forecast the future movements in the market prices of different shares. Generally, there are two methods to do so.

Technical analysis explores the underlying principles of the movements on the market. The analyst does not need to know anything about the firm which issued the shares. The analysis forecasts the changes and developments in the market prices built exclusively on the following factors: change in the market prices, trading intensity (increasing or decreasing), other calculated indicators, and the behaviour of other participants on the market. The predictions are based on the different aspects of market price charts. They can be used for short-, medium-, and long-term trading.

Fundamental analysis examines the cause of the behaviour of the market. It attempts to forecast the changes in the share prices based on the following factors: the economic indicators of the given company, the assumed future success of its products, the assumed expertise of its management, and the political (or other) environment influencing the firm's operations. This type of analysis is primarily used for long-term investments with maturity of several years. For short-term investments (such as day trading) only the current effect of changes in factors determining the value of the firm can be used (e.g. quarterly reports, other news).

1.5. Risk and diversification

The value of an investment depends to a great extent on the return and the risk associated with the given investment. If investors think rationally, they will expect higher returns for riskier investments. Thus, investments with higher risk must offer larger returns. This return is the risk premium.

According to the modern portfolio theory, investors should divide their money – designated for investments – to minimize risk. This practice is called diversification. The smaller the correlation between the movements of two shares, the smaller the overall risk of the portfolio. On the stock exchange, diversification can be achieved geographically (between shares from different countries), or industrially. It is also possible to diversify a portfolio with shares of export-oriented companies or rather with shares of companies depending on their domestic market. The risk of a perfectly diversified portfolio equals to the country risk of the given market. Hence, the ultimate goal of collecting risky shares is to make the overall portfolio risk smaller than the individual risks of the different shares in the portfolio.

Risk aversion is an ordinary human behaviour. However, it usually restricts the investor from possible large gains on the market. When someone wishes to earn more money on the exchange market, risk aversion and profit orientation narrow down the possibilities, because everyone intends to maximise return and minimise risks. This risk aversion indicates that (because of the short-term thinking) investments with high long-term returns are not considered, if their short-term return is volatile. It is right to say that **the market is driven by the combination of hunger for profit and loss avoidance.**

1.6. Earnings

One of the most important regulations concerning firms on the stock exchange is the mandatory publication of quarterly reports (earnings). These reports have similar function as the yearly financial statements, except they are simplified. All data in the reports must be real in terms of accounting standards. The aim is to show the public how much profit (or loss) the company has achieved in the period of 3 months, and what are the predictions for the upcoming quarter. The forecasts can influence the price of the share to a great extent.

Earnings are published on a predetermined date. Firms also indicate whether the announcement is going to happen before the stock exchange opens, during its operations, or after the day closing. Sometimes companies even communicate the precise time of the publication within the opening hours of the stock exchange.

The most important data in the report is the net profit. Before announcement, an expected value is calculated based on the objectives of the firm and on the estimations of several market analysts. The difference between the expected and the actual value may result dramatic changes in the market price of shares, based on how well the company could achieve its goals and execute its plans in the quarter. Many business analysts can also predict whether companies are going to meet their short-term goals. Therefore, unexpected changes can happen even before the publication of the report. When prices start moving rapidly in one direction, many investors may think this movement is going to be a significant trend. Because of that, they will follow the direction of the market, further increasing the price change. This situation is called herd behaviour, and it can be an alarming event for all investors. If the news is found to be fake, these investors will leave their positions suddenly, resulting the prices to change drastically in the other direction. This phenomenon can happen repeatedly, causing huge fluctuations in the price.

The earnings can be handled easily – just like the fundamental items – because only those shares can cause problems, which are announced during opening hours. Because people know this information in advance, they will not trade on that day or will handle the shares even more cautiously than usual.

1.7. Initial Public Offering (IPO)

Going public is a significant step in every firm's life. Country specific accounting standards and regulations determine the life of all companies, listed or not listed on the stock exchange. In the case of public companies, the rules of the country's stock exchange must be met as well (apart from the laws and the accounting regulations concerning ALL businesses). This means more publicity, transparency, and obligation to inform the public and the shareholders about company related issues.

The process of going public is implemented by businesses specialised in these procedures (investment banks). For an IPO, the following two fundamental steps must be executed:

- The firm's operations (accounting, decision making, general business regulations, etc) must be reorganised to comply with the stock exchange regulations.
- The value of the firm must be estimated realistically, as the opening price of the share is based on the valuation.

The latter is especially important in this course, because the appearance of a new firm on the stock exchange is always an interesting event. The trading of the new shares starts at a pre-specified time, during the opening hours of the exchange. This is element is fundamental and easy to handle, because the lack of information protects the players on the market from trouble.

Moments after the IPO, two scenarios are possible. One is when **the trading starts with huge interest** in the new shares and within minutes, millions of shares circulate between the market participants. In this case, the price of the share starts rising rapidly. The share price will also fluctuate: compared to the initial price, the pike during the day will be considerably higher. At the end of the day, the closing price will end up somewhere between these two prices. The most famous IPO in the history of NASDAQ happened with BIDU: the initial price of a share was $65 and within two hours, the price climbed up to $150.

The other scenario is when **the market shows small interest in the new shares.** There will be small number of deals with the share and the closing price will stay close to the initial price. Big drops in share prices immediately after IPO happen rarely, as it would indicate the lack of expertise of the firm executing the IPO (regarding the valuation of the company). In the United States, it is even possible to ask for compensation if this happens.

1.8. Online trading

For the first time in the history, there is an opportunity to manage bank transactions, contact brokers, and track the current market prices of shares online. Simply put, anyone can **invest in companies operating anywhere around the world directly from home** and can monitor the current state of all his or her positions online.

When investing private savings in shares from home, one can become a co-owner of a public company residing in any country. Despite any geographical distance, the new owner of the share(s) is part of the future of the firm (from the time of the purchase), thus anyone can access a powerful position. With the spread of Internet access, it is possible for everyone with a computer to buy and sell shares easily from home.

The online trading is based on the co-operation between different countries. The basis of the collaboration is the settlement system between stock exchanges, which makes it possible to buy foreign shares from local brokers legally, quickly, and easily.

The sale and purchase of financial instruments within the same trading day is called day trade. Day trade is when someone buys and then sells a share on the very same day, or when first the sale and then repurchase happens (on the same day as well). The only difference between the two is whether one has cash or possesses a share at the beginning of the trading. In the case of day trades, brokerages usually charge less commission than normally. Therefore, even a daily increase of 1-1.5% can result a more significant income. It is not mandatory to close an open day trade until the end of the trading day. However, if the day trade is not closed, the brokerage firm will charge the normal commission, as the transaction then counts as normal selling or buying on the stock exchange market. In Hungary, day trades can be executed with any shares listed on the Budapest Stock Exchange.

1.9. Market news

The market theory states that news is moving the market prices through investment decisions. However, there are huge differences between two news from the aspect of trustworthiness and their public announcement. The two opposite ends of the scale are the official announcement of facts and the spreading of rumours.

Individuals organised into groups based on a common interest can be easily manipulated and influenced. Manipulation is further intensified when information comes from people, groups, or institutes with authority and reliability on the market. Furthermore, media plays a significant role in transmitting the news. Due to the increased speed and scope of the spread of information, stock exchange processes are changing more rapidly, and amplitudes are stronger now than ever before.

The main problem is that nobody is able to determine whether the news will have positive or negative impact on the price of the firm's shares. One part of the investors may think it is good news and long on their positions. In the same time, other investors may interpret the very same information as bad news and go short. This situation leads to big jumps in the market prices in both directions, and the movement of the share price becomes totally unpredictable. The stock exchange may suspend all trading with the given share in order to protect investors if one piece of news may induce a huge fluctuation in market prices of a share. This practice usually happens when specific bad news concerning a firm leak and the impact is not obvious or would not grant enough time for the firm to react to the news officially in a public announcement.

The exact time of the “birth” of a narrow group of news can be known in advance. These have the same impact as unexpected news, but at least the firm has the chance to prepare for it. Examples include when a firm announces a strategic decision influencing its future, the announcement of a new product, or other business plans announced publicly.

As the famous phrase says: **“buy the rumour and sell the news”**.

2. Technical analysis

2.1. Patterns

When connecting the peaks and throughs of the moving market prices, one can discover geometric formations. Officially, these forms are called chart patterns. Patterns carry reliable information for trading, because they look at the big picture and suggest information about the future behaviour of the prices with high punctuality. Trading volume has a significant role in the interpretation of the patterns, as volume strengthens the role of patterns.

Patterns can be divided into two categories:

- continuations,
- reversals.

Continuations signal that the trend will continue once the pattern is complete. Therefore, opening long positions is highly recommended in this pattern.

On the contrary, **reversals** indicate that the trend will reverse upon completion of the pattern. In this case, it is suggested to close the long and open short positions. Patterns cannot be interpreted in long term. In longer time horizon, only trends can be taken into account. During technical analysis, one can meet patterns in both medium and short term.

2.2. Charts

There are different **types of charts** to illustrate market prices. There are four different chart types introduced in this lesson.

Line Chart

Some traders and investors prefer to see the closing prices over seeing opening, minimum or maximum prices. This type of illustration is possible to use even if there are no other data available than the closing prices. The line chart is drawn simply by **connecting the closing prices.**

Line chart

Open-High-Low-Close (OHCL) Chart

OHLC chart is a type of bar chart. It shows the minimum, the maximum, the opening, and also the closing prices. The **opening and the closing prices are indicated by horizontal lines.**

OHCL Chart

Candlestick Chart

The first use of this chart type dates back to 300 years ago in Japan. Currently this type of illustration is quite popular. The **minimum, maximum, closing, and opening prices** are easily observable on candlestick charts.

Candlestick Chart

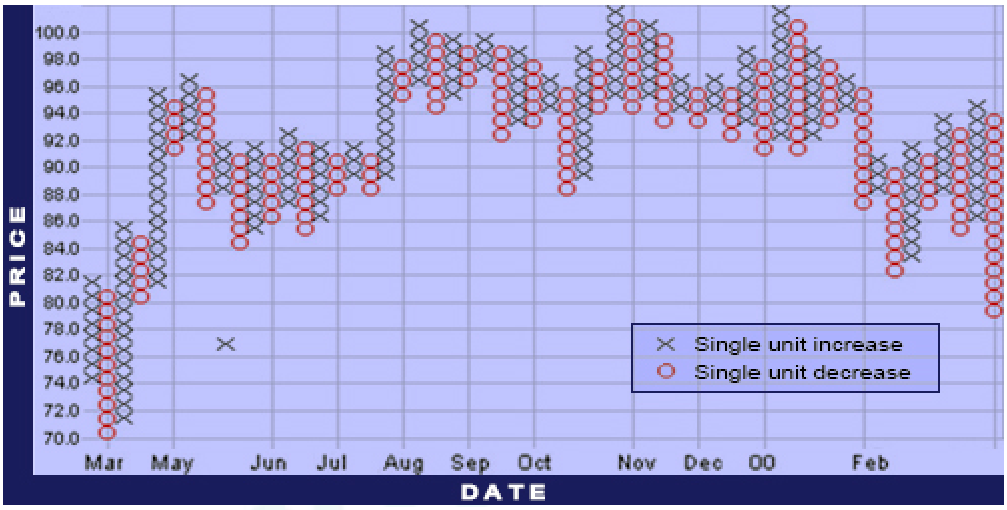
The interpretation of candlestick figures is the following:

- **Green, filled bar:** The current day's opening price is higher than the closing price on the same day. The previous day's closing price is lower than today's closing price.
- **Green, empty bar:** The current day's opening price is lower than the closing price on the same day. The previous day's closing price is lower than today's closing price.
- **Red, filled bar:** The current day's opening price is higher than the closing price on the same day. The previous day's closing price is higher than today's closing price.
- **Red, empty bar:** The current day's opening price is lower than the closing price on the same day. The previous day's closing price is higher than today's closing price.

The vertical lines above and below of the body of the candlestick indicate the maximum and the minimum prices. They are called the candlestick's shadows.

Point & Figure Chart

The popularity of the point and figure chart is in its simplicity. **Every point illustrates a given time period.** Thus, this chart shows the price changes and eliminates the time factor (there is no time axis, only a price axis). Every X or O takes a up a box on the chart. Every chart has a specific box size. The size tells how much the price has to increase (X) above or decrease (0) below the current box to make it to the new box. Every chart has a reversal amount as well. It shows how much the price has to move in the opposite direction before turning backwards. Before reaching the reversal point, a new column starts before the previous one, in the opposite direction. Small movements in the market price or still market prices do not result more figures on the chart.



Point & Figure Chart

To summarise: **O symbolises falling prices, and X symbolises increasing prices.** For example, $1 setting is frequently used for $20 or more valuable shares. This simply means that a 0 is drawn when the price of the share falls with $1. If it increases with $1, an X is going to be drawn.

The illustration helps to recognise support and resistance lines, and also breakouts.

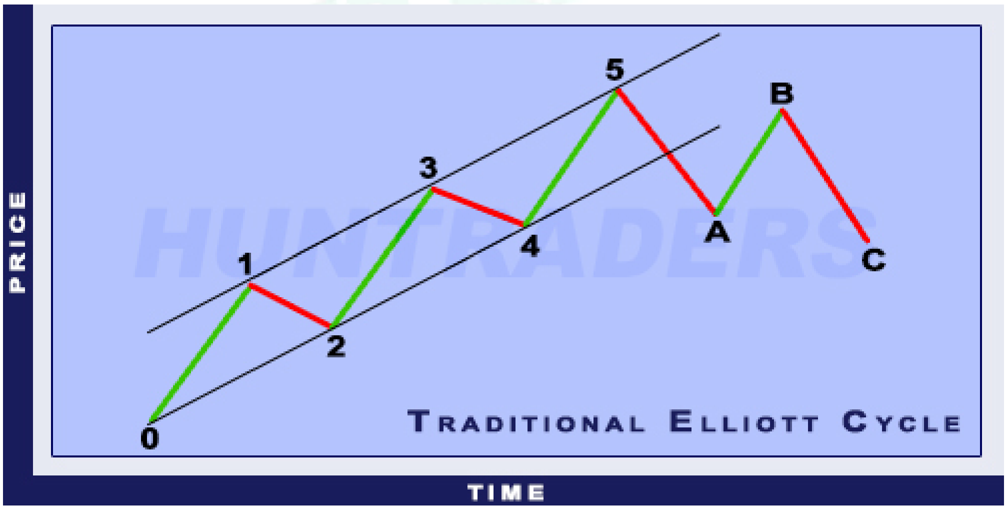
2.3. Cycles

Market prices of shares in a given period can show cyclical tendencies. Cycles are made up by an upward trend, immediately followed by a downward trend. Cycles are formed by primary cycles which develop in longer term. Within them, secondary cycles are found. Primary cycles are usually built up by five secondary cycles. Three of these secondary cycles have the same direction, the other two are in the opposite direction.

In case of an upward trend, the support line is drawn by connecting the throughs (minimum points). Connecting the peaks (maximum points) gives us the resistance or channel line. Whereas, in case of a downward trend it is the opposite: the support line is drawn by connecting the peaks and the resistance line is drawn by connecting the minimum points. A trend is upward, if the peaks are getting higher and higher. A trend is declining if the throughs are getting lower over time. When the market price goes outside the channel (resistance and support lines) and stays outside by the end of the day, it is called breakout. If the breakthrough does not exceed 2%, it is called a “dead” breakout.

Elliott cycle

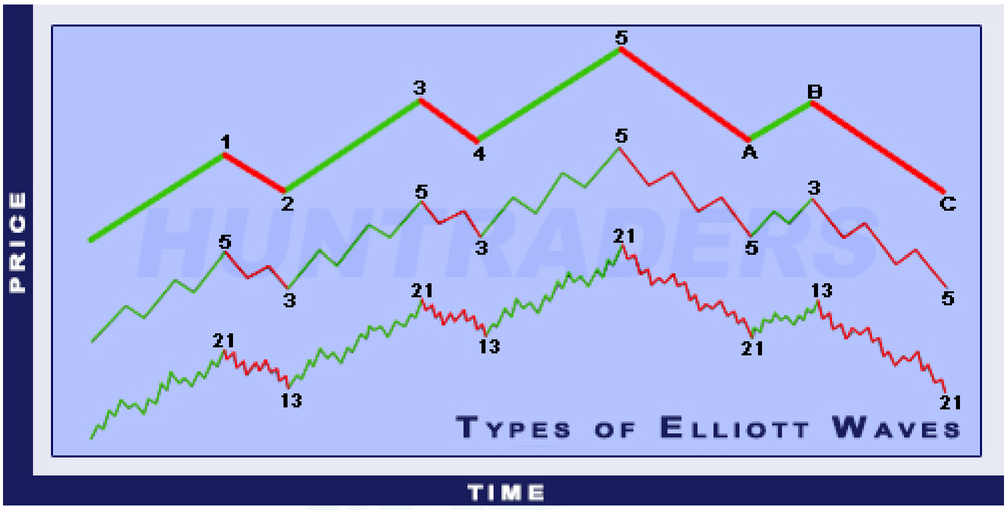
Markets have wavelike movements, according to the analysis method based on the Elliott Wave Theory. Movements have two main phases: impulsive and corrective waves. The impulsive phase contains 5, the corrective phase contains 3 wave sections. The peaks of the impulsive phase’s waves are always higher than the previous peaks, until the corrective phase starts. Elliott waves behave fractally, meaning that more smaller Elliott waves can add up to a big one. Therefore, waves can be analysed in short-, medium-, and long-term as well.



Traditional Elliott Cycle

The Elliott Wave Theory distinguishes between 9 types of waves based on their size:

- Grand Supercycle (multi-centrury)
- Supercycle (multi-decade)
- Cycle (one year to several years)
- Primary (few months to a few years)
- Intermediate (weeks to months)
- Minor (weeks)
- Minute (days)
- Minuette (hours)
- Sub-Minuette (minutes)



Types of Elliott Waves

Elliott Wave Theory works the following way

- 1st wave: The appearance of the first wave does not immediately mean the beginning of an Elliott Wave. When the first wave appears, the news on the market are usually not optimistic. The previous trend is still in effect. Analysts are occupied with the calculation of low returns and the economy is still considerably weak. The market is still bearish, put options are frequent, and the options market is characterised by high implicated volatility. Because of the rising market prices, trading volume will slightly increase.
- 2nd wave: The second wave is a correction, but it never falls below the starting point of the first wave. The news on the market are still pessimistic. As the market prices drop, the decreasing market perception will take place. However, when investigating the situation deeper, one can see positive signs: the trading volume is smaller in the second wave and the prices do not drop below 61.8% of the first wave. To put in other words: the correction retrieves 1/3 of the impulsive phase.
- 3rd wave: The third wave is usually the largest and the most powerful. Market news turn positive and the analysts increase the value of predicted returns. The market price grows rapidly, the correction period is really short. At the starting point of the third wave, one can meet negative news. Most of the investors talk about decreasing market considerations at this time. Around the middle of the wave, investors realise the fact that a new upward trend has started. The ratio between the first and the third waves is usually 1.618:1. From this point, one can talk about the development of an Elliott Wave, because the basic criteria are met.

- 4th wave: It is the correction of the previous, impulsive wave. This wave shall never drop to the level of the first wave. The trading volume will stay considerably lower than the previous wave's. The length of the third and fourth waves are the same. However, the length of the second and fourth wave can differ.
- 5th wave: The fifth wave is the last leg of the trend. The market is bullish and the market news are positive. This is the point when most of the investors buy. The trading volume is lower than it was in the third wave and most of the momentum indicators show divergences (the price reaches new peaks, while the markers of the indicators drops to new minimums).
- A wave: The market price breaks out of the channel. The news during the A wave are still optimistic. Most analysts think that it is a smaller correction of the upward trend. Some of the technical indicators move along with the wave. The A wave is characterised by increased trading volume, while the option market is characterised by increasing implicated volatility and an increasing number of open positions.
- B wave: The B wave is the correction of the previous wave. It regains its 50 or 61.8%. It may seem like the beginning of a new, long-term upward trend. Investors who are proficient users of technical analysis will recognise the right shoulder of the reversal head-and-shoulders. The trading volume is going to be lower than in the A wave. From this point, economic indicators do not increase further, but also do not reverse.
- C wave: In the C wave, the market prices drop sharply. In the same time, the trading volume increases. Around the 3rd part of the C wave almost everyone realises that a new downward trend has started. Usually the C wave has the same length as the A wave, but it can sometimes grow 1.618 times longer than the A wave. When the C wave drops below the A wave, the Elliott Wave is considered to be closed.

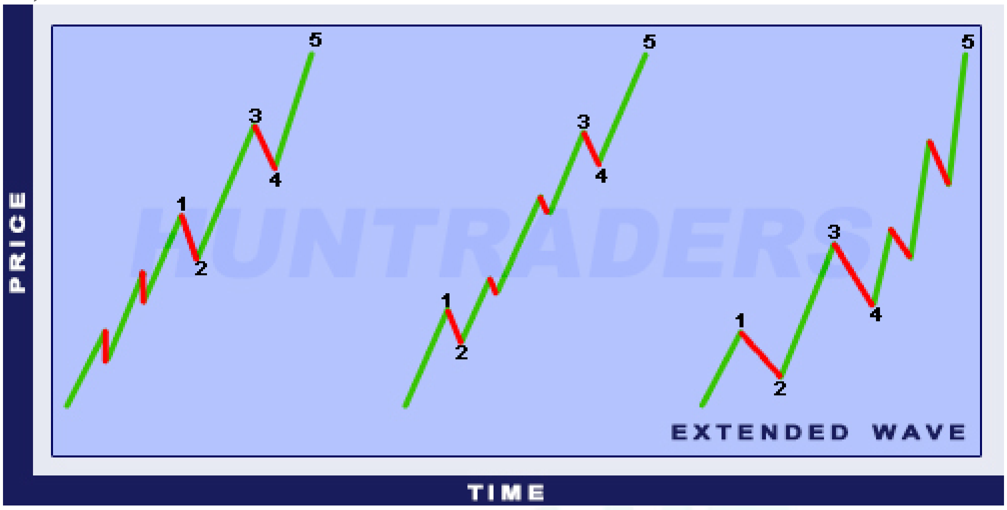
Trading signals

The first two waves of the Elliott Wave can be similar to a simple continuation pattern. If the second wave does not fall below the first wave, and the time period of the first two waves equal to the time period of the third, then the time of the breakout can be predicted with high punctuality. Then a long position can be opened. The breakout usually comes with high trading volume and usually a gap. Approaching the end of the 5th wave, a trading signal may be received. At this time the trading volume increases and then it suddenly stops. Reversals are signalled differently, but they need 2-3 weeks at least to develop.

Upward phase

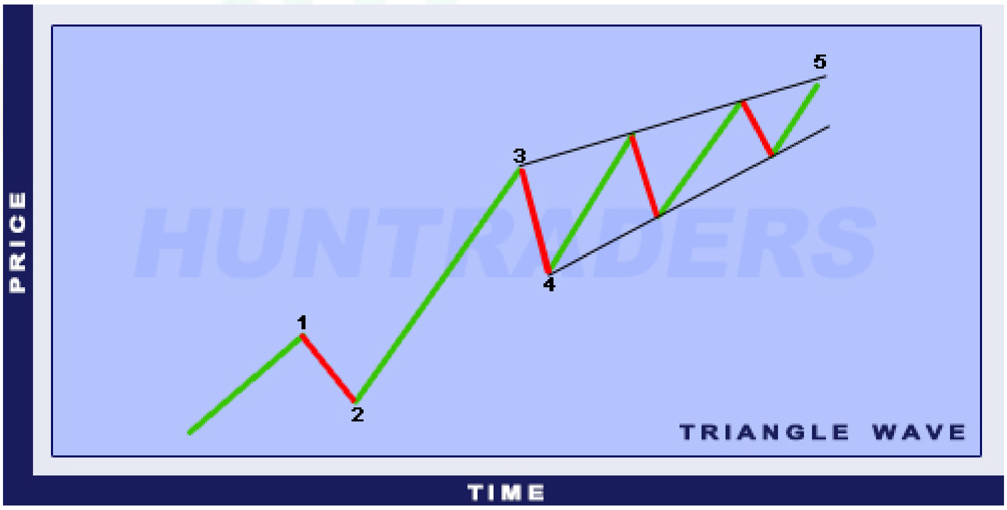
Upward phases do not always show a regular Elliott Wave, however the first three waves are developed in the usual way in 80-90% of the cases. Irregularity has three common types.

1. Extended Wave. Only the 1st, 3rd, and 5th waves can be more extended than the others. Extended means that these waves are prolonged. Thus, these waves are built up by other, smaller waves.



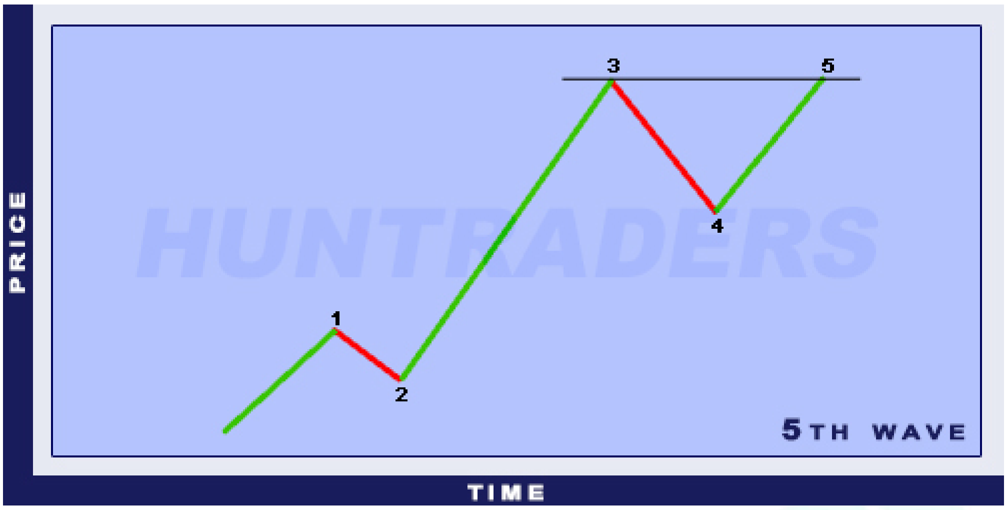
Elliott Extended Wave

2. Diagonal Triangle at Wave 5. Sometimes the momentum of the 5th wave is as weak as the 2nd and 3rd waves altogether. Then the 5th wave “builds up” a triangle.



Elliott Diagonal Triangle Wave

3. 5th Wave Failure. Under certain circumstances, the 5th wave can be so weak that it does not even reaches the top of the third wave. Then it forms a double top pattern at the peak of the Elliott Wave.

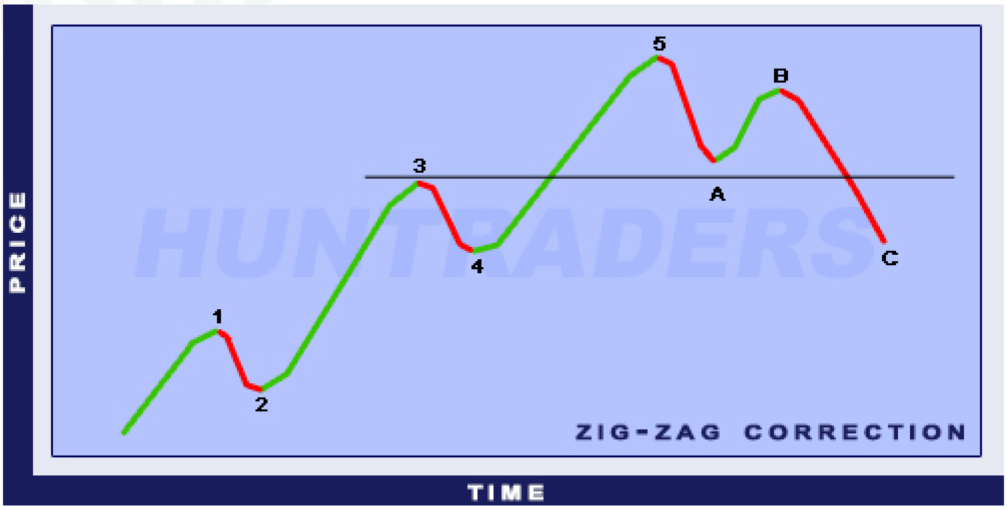


Elliott 5th Wave Failure

Correction phase

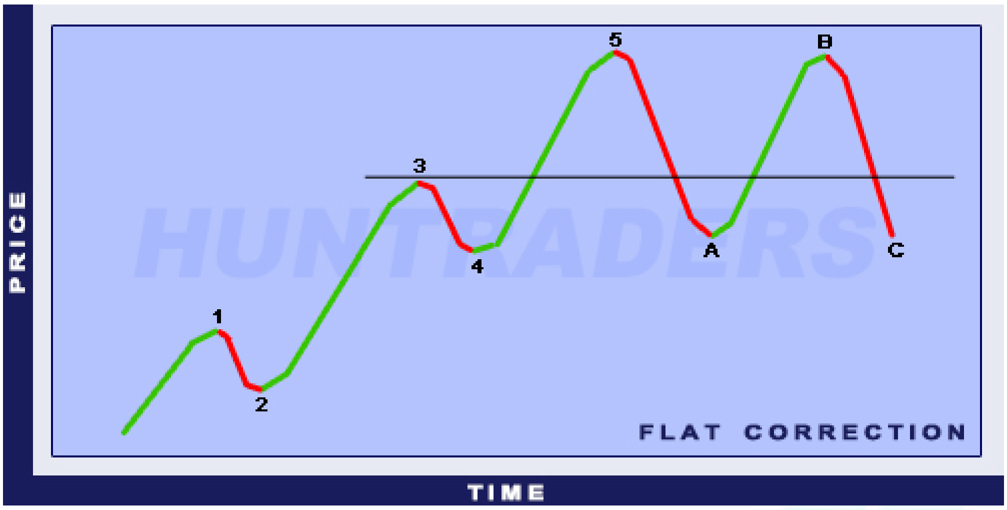
Trading with Elliott Waves is difficult, because the pattern can be only recognised only in 60% of the cases. In the rest of the cases the correction phase differs from the regular pattern.

1. Zig-zag. The A point does not fall below the 3rd wave. Furthermore, the B wave takes back 50-75% of the A wave. The rest of the wave is regular. The C wave equals to the A wave 1.618 or 2.618 times.



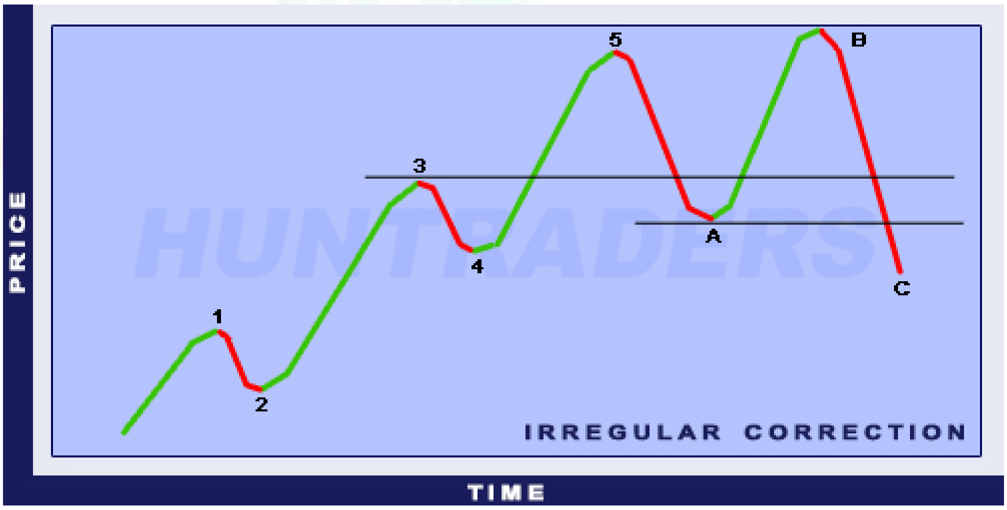
Elliott Zig-Zag Correction

2. Flat. The A wave is regular. The correction of the B wave extends to the top of the A wave. Therefore, the C wave will not reach a new minimum point. It will stop at the A wave's minimum.



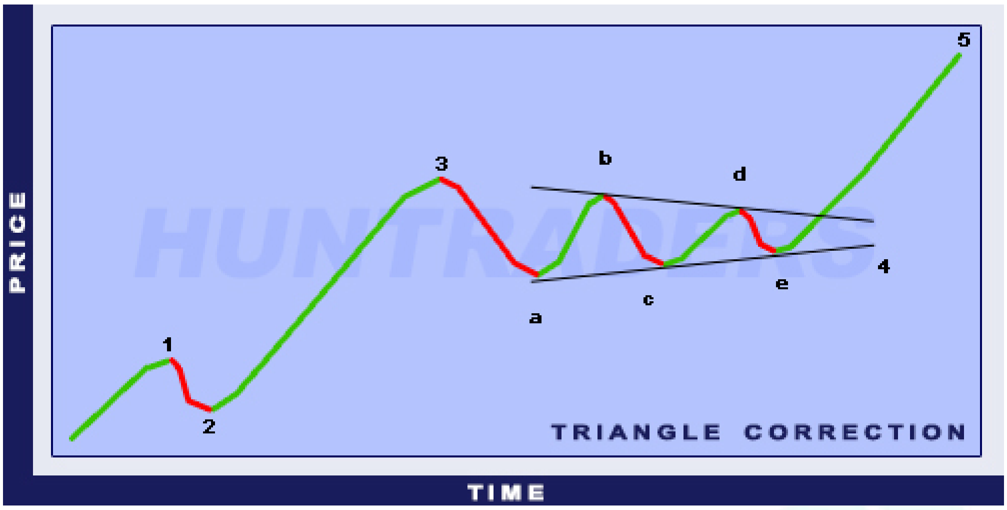
Elliott Flat Correction

3. Irregular. This correction is the most irregular. The B wave's peak is higher than the maximum point of the 5th wave. Usually there is no increased trading volume at the B wave's top and the raise stops quickly. Then the C wave drops below the A wave's minimum point. The B wave is 1,5 or 1.25 times the size of the A wave. The C wave is 1.62 or 2.62 times the size of the A wave.



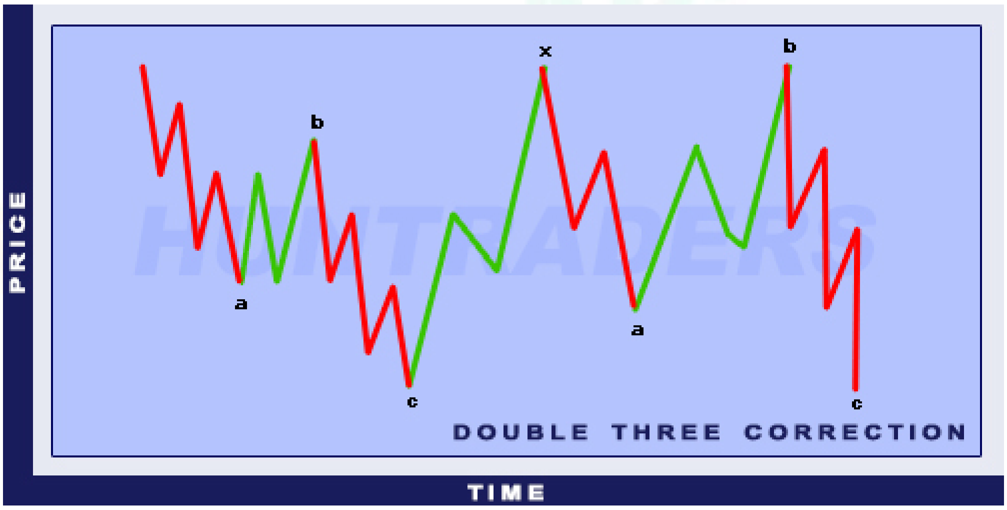
Elliott Irregular Correction

4. Triangle. This correction pattern is made up by 5 smaller waves. These waves are the A, B, C, D, and E waves, and they make up a triangle. The breakout point is at the breakout of the B-D line's breakout, which is below the peak of the C wave.



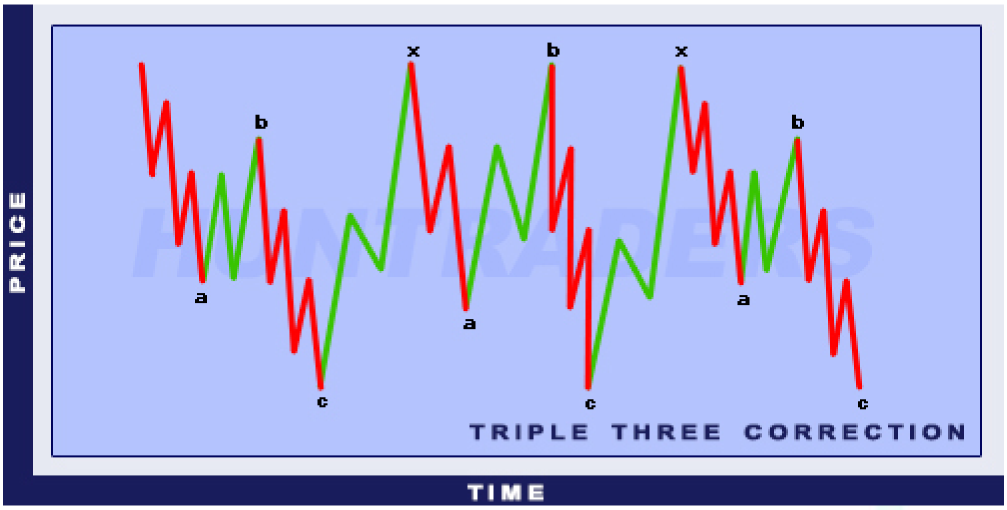
Elliott Triangle Correction

5. Double Three. This type of correction is made up by two of the above mentioned 4, which are separated by an X wave from each other.



Elliott Double Three Correction

6. Triple Three. This type of correction is made up by 3 of the above mentioned 4 corrections. They are also separated by an X wave from each other.



Elliott Triple Three Correction

2.4. Channels

In most of the cases, prices move in channels. To show the channel line, a straight line is drawn parallel with the trendline, until the minimums and maximums. In case of an upward trend, the channel line acts as the resistance. On the other hand, when a downward trend develops, the channel line will act as the support. When a massive breakout (with large volume) occurs, it is assumed the trend will probably continue. The scale and the direction is highly predictable.

- The scale of the increase/ decrease equals the width of the channel
- The direction of the movement is the opposite of the direction of the breakout

The role of support and resistance lines will be inverted after a breakout in case of a horizontal trend. This phenomenon can be observed for upward and downward trends as well.

Channels

1. **Trendline:** At least two values are required to draw a trendline. The trendline clearly defines the direction of the trend. It is going to increase for an increasing channel, and will contain at least two minimum values. On the contrary, for a decreasing channel, the trendline will decrease and contain at least two maximum values.
2. **Channel line:** Channel line is parallel to the trendline. In ideal cases, the channel line is given by two maximum or minimum values. However, there may be cases when only one value is available and the channel needs to be drawn based on the trendline.
3. **Upward channel:** The trend is upwards until the market price moves and increases within the channel. The first sign of a change in a trend is when the market price falls below the support line, outside the channel. The next similar event increases the probability of a trend change. Market prices above the resistance line indicate the acceleration of the trend.
4. **Downward channel:** The trend is downwards until the market price moves and decreases within the channel. The first sign of change in a trend is when the market price rises above the resistance line, outside the channel. The next similar event increases the probability of a trend change. Market prices below the support line indicate the acceleration of the trend.

Some resistance and support lines have greater importance than the others. **The more of the following criteria the lines meet the stronger they are:**

- The more times the price reaches the channel the more important it is
- The bigger the turnover near the channel line the more important it is
- Newly formed lines are more important
- The wider the channel the more important it is

Technical analysis is hardly ever accurate. Because of that, **resistance and support zones** are defined instead of lines. The width of the zone is usually 2%. When the closing price is outside the zone, a breakout happens. Movements within the zone are rather interpreted as warnings.

Support and resistance zones

2.5. Candlestick forms

The origin of candles

Technical analysis was first used in Japanese rice trading in the XVII. Century. Despite the several differences, Charles Dow theory (early 1900s) has fundamental similarities with it:

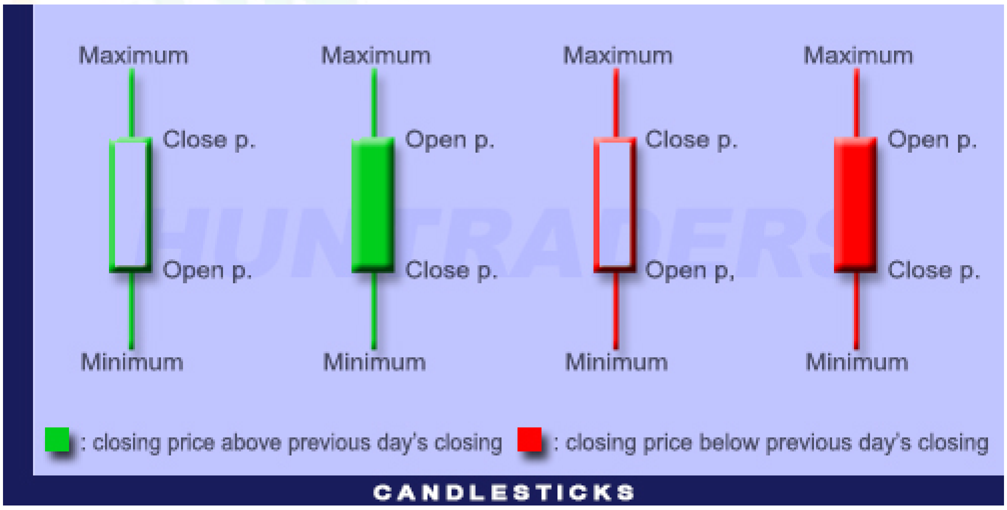
- "*What*"-s are more important than "*why*"-s (news, quarterly reports, etc) for the development of the market price.
- Every known information is incorporated in the market price.
- The sellers and buyers move the market based on their expectations and feelings (fear and greed).
- This market is unsteady.
- The current market price does not reflect the actual price (or value) of the underlying asset.

Candlestick visualisation has reappeared in the 1850s and has not stopped spreading ever since. The knowledge, the usage, and the correct interpretation of this chart type has changed and developed over the years.

Formation

One must know the opening, closing, maximum, and minimum values to draw the financial instrument for the analysed period. The core of the candle (whether filled or empty) is called the body. The lines above and below the candle are called shadows.

If the closing price is above the opening price, the top line of the body will show the closing price. If the closing price is below the opening price, the top line of the body will show the opening price.



Illustrating the opening, closing, minimum, and maximum values with candlesticks

Compared to other traditional line charts, candlestick visualisation tells more about the movements in the prices. The opening and closing price is directly and easily observable and comparable with the previous time periods. Candlesticks with empty bodies show that the closing price is above the opening price. They indicate a buying pressure. Candlesticks with filled bodies show that the closing price is below the opening price. They show a selling pressure.

Consecutive, individual candlesticks form groups or formations. The development of candlestick formations may take several years. They visualise the investor's behaviour. Despite signalling a buy or sell action, candlesticks should not be used to trade without applying other measurements of technical analysis as well.

2.6. Indicators

What is an indicator?

Indicators are a series of numbers, whose value is derived from generic price activity in a financial instrument. The prices used for the calculation are usually a combination of the opening, closing, minimum and maximum prices for a given period. Some indicators rely solely on closing data, while others are using market prices of current transactions for the calculations. Thus, we obtain indicators with the help of different mathematical algorithms of price data.

For instance, the average of three closing prices may be included in the dataset the following way: [(22+25+25)/3=24]. However, the value of the indicators do not provide additional information about a stock, it only serves as a basis for the technical analysis. By arranging the data in a sequence, the past behaviour of the stock price can be compared with its present behaviour. **The indicators are usually displayed above or below the graph showing the market prices.** As soon as the price and indicator graphs are observable side by side, their movements are comparable. Sometimes indicators are visualised on the market price graphs for easier comparability.

What do indicators offer?

Technical indicators analyse the price changes from several aspects. Some indicators, for example moving averages, are based on simple mathematical formulas and are easy to understand. Other indicators, such as the Stochastic Oscillators, are built on complex mathematical formulas and it takes time and practice to understand. By the complexity of the formulas, indicators offer more perspectives to the direction and the intensity of the instrument.

Simple Moving Average (SMA) is an indicator calculating the average of the market price of the stock for a given period. If the share is volatile, the moving averages help to smooth the dataset. They filter out the random noises and spikes on the curves and give a finer image about the price changes. XYZ share is highly volatile, therefore the price can jump upwards or downwards suddenly. Using a daily SMA, the fluctuations and spikes can be smooth out and the trends can be observed more easily.

Why are indicators uses for technical analysis?

Indicators offer three functions: warning, confirming, predicting.

- Indicators warn traders. If the momentum indicator is declining, it may signal a breakout. On the other hand, the positive divergence between the share and the indicator may result the breakout on the resistance line.
- Indicators confirm events signalled by other technical analysis tools (candlesticks, chart patterns). When the price breaks out of the channel, the SMAs' intersections can confirm the breakout.
- Some analysts use the indicators to predict future movements in the share prices.

Advices for using indicators

Indicators send signals. This statement may sound simple, but sometimes traders ignore the movement of market prices and focus only on the movements and changes of indicators. Indicators describe the movement of prices with the help of mathematical formulas. Hence, they rather reflect the derived data instead of the changes of the market prices. This should be taken into account during the analysis.

Despite the obvious signals to sell or buy, indicators should be relied only with the use of other technical analysis tools as well. It can happen that the indicator suggests buying, but the chart shows the development of a decreasing triangle. In this case the indicator shows a false trade signal.

Interpreting the indicators and their signals is more like “a form of art” than real science. Some indicators behaviour may differ for different shares. Indicators working accurately for IBM may give false signals for Google. One must pay attention during analysis and familiarise with the use of more indicators. The more familiar one gets with more indicators the easier it becomes to interpret them.

Currently there are hundreds of indicators used, and the pool of indicators is continuously expanding. **Out of the hundreds of indicators, only a few can provide truly meaningful information.** These include the moving averages, the MACD, the RSI, and the Stochastic Oscillator for instance.

It is recommended to be careful when choosing an indicator for technical analysis. **It is useless to use more than 5 types of indicators.** The combination of two or maybe three indicators is the ideal. **Complementary indicators are suggested to be chosen. Using indicators generating similar signals is not recommended.** For example, it is unnecessary to use two indicators that are showing overbought/ oversold levels (such as the Stochastic and RSI). These indicators are momentum types. One should use different types of indicators.

2.7. Gaps

Gaps on the market curves are also patterns, carrying important and useful information. **Gaps are empty spaces between two consecutive trading periods.** In case of an upward trend, a gap indicates that the next day’s minimum is going to stay above the maximum of the previous day. If there is a downward trend, the opposite will happen. According to the principals of the technical analysis, the gaps are going to disappear due to basic market mechanisms. Gaps are observable on a daily basis. Weekly or monthly trends are rather rare. Gaps can last 1-2 days, a week, or even several months, and sometimes it happens when gaps are not filled. Gaps indicate the beginning, the end, or the widening of a process.

There are four categories of gaps:

- Common gap
- Breakaway gap
- Runaway (or measuring) gap
- Exhaustion gap

2.8. Trends

The market prices of shares are usually following trends. One can distinguish between short-, medium-, and long-term trends. Long-term trends are shaped by market expectations, while the short-term ones are shaped by current news and events. Trends are illustrated on charts.

There are three main types of trends:

- Upward trend line (ascending trend)
- Downward trend line (descending trend)
- Horizontal trend line

In case of an upward trend, the support line is drawn by connecting the throughs (minimum points). Connecting the peaks (maximum points) gives us the resistance line. Whereas, in case of a downward trend it is the opposite: the support line is drawn by connecting the peaks and the resistance line is drawn by connecting the minimum points. A trend is upward, if the peaks are getting higher and higher. A trend is declining if the throughs are getting lower over time. When the market price goes outside the channel (resistance and support lines) and stays outside by the end of the day, it is called breakout. If the breakthrough does not exceed 2%, it is called a "dead" breakout.

Upward trend line (ascending trend)

Upward trend line (ascending trend)

- drawn by connecting successive minimums,
- the more minimum points there are, the stronger the trend,
- also called the support line.

Downward trend line (descending trend)

Downward trend line (descending trend)

- drawn by connecting successive maximums,
- the more maximum points there are, the stronger the trend,
- also called the resistance line.

Horizontal trend line

Horizontal trend

- drawn by connecting successive minimums or maximums,
- the more minimums or maximums points there are, the stronger the trend,
- called support line if the market price is above the trendline and called resistance line if the price is below the trendline.

3. Fundamental analysis

3.1. Fundamental analysis

The fundamental analysis **examines the underlying causes of the market's behaviour**. It attempts to forecast the changes in the stock prices based on the following factors: the economic indicators of the given company, the assumed future success of its products, the assumed expertise of the management, and the political (or other relevant) environmental factors influencing the firm's operations.

Through analysis, the desired information is obtained directly from the financial statements of the firm. All decisions related to the market price of the company are based on these statements. The share may be beneficial to buy if these data seem to be favourable compared to the current price of the stock. On the other hand, if the firm is overpriced on the market then the owner of the share should consider the opportunity to sell.

The two most important financial statements are the **balance sheet and the income statement** (also called profit and loss (P&L) statement):

- The balance sheet captures the current state of the firm. It shows where the capital came from (owner's equity and liability side) and how it has been used or invested (asset side).
- The income statement shows what happened in a given period (quarterly, semi-annually, or annually): the turnover (revenue) of the firm, the expenses of the period, and the profit earned at the end of the reporting period.

3.2. Market indicators

When trading stock, one may see market indicators the most frequently, because these ratios can give a quick yet comprehensive overview of the firm assessed. Market indicators involve constantly changing market prices. Hence the ratios follow the movements of the price of the shares.

Market indicator types:

- Market capitalisation
- Earnings per Share (EPS)
- Price per Earnings (P/E)
- Price per Book Value (P/BV)
- Dividend yield

Market capitalisation

Market capitalisation is the number of the company's shares multiplied by the actual price of the share on the stock exchange. In other words, it shows the **valuation of the company by the group of shareholders.**

Earnings per Share (EPS)

Earnings of the company is publicly available data, accessible in company-related materials (such as the quarterly or yearly financial reports). EPS is calculated as the earnings divided by the number of outstanding shares on the market. By looking at the EPS, one can see **value of earnings generated by the firm per one piece of share**.

Price per Earnings (P/E)

P/E is the ratio of the actual stock price and the EPS. When calculating P/E, usually the next (or the upcoming) period's forecasted EPS is used. The EPS forecasts can be obtained from the firm itself or from the continuous forecasts made by the analysts of brokerages. Simply put, P/E is a ratio, indicating **how many times higher is the price of the share compared to the earnings** of that one share. From this number, any trader should be able to assess whether the stock is under- or overpriced in that moment.

Price per Book Value (P/BV)

If all assets are sold and then all debts of the firm are paid, the remaining amount of money is the property of the shareholders. The market value of the shares are usually higher than the book value of the shares. However, it is up to the trader to decide from what ratio is it worth to invest in a company. If the P/BV ratio is smaller than 1, then the firm is under-priced, as the market value of the share is less than the value of the company's assets.

Dividend yield

Dividend yield is the ratio of the expected dividend per share and its market value. Corporations usually have predictable dividend policies: they only distribute a given percentage of the net profits as dividends. With this practice, firms assign further capital to further expansion.

3.3. Profitability indicators

The indicators in this chapter quantify the profitability and the returns achieved by the company.

Profitability indicator types:

- Return on Equity (ROE)
- Return on Asset (ROA)
- Return on Sales (ROS)
- Coverage indicator

Return on Equity (ROE)

ROE shows how well managers are employing the funds invested by the firm's shareholders. It is the **net profit of the company divided by the shareholders' equity** shown in percentage.

Return on Asset (ROA)

ROA **compares the size of the total assets with the net profit** achieved by these assets, therefore it indicates how much profit the company is able to make for each euro of asset invested.

Return on Sales (ROS)

ROS is also called the net profit margin and it shows **how much the company is able to keep as profits for each euro of sales** it makes. The result of this ratio is relative: average values vary in different sectors, industries. For instance, a company offering services can be more profitable in this term than a "traditional" commercial firm.

Coverage indicator

The coverage ratio shows the relationship between the coverage (*difference between revenues and overheads*) and the revenues. This ratio also greatly depends on the scope of activities of the firm.

3.4. Efficiency indicators

Efficiency indicator types:

- Inventory turnover
- Trade receivables turnover
- Trade payables turnover
- Other industry-specific indicators

Inventory turnover

Inventory turnover shows how many times the firm has **sold out its inventory in a given period**. The nature of the industry should be considered during the assessment of the value, because a large inventory turnover may arise from a relatively small size of average inventory. Thus, bigger turnover does not necessarily mean better performance in every case. It is calculated as sales divided by *average* inventory of the period.

Trade receivables turnover

This ratio shows **how efficiently the firm collects its receivables** from its customers. More precisely, the number of times the business has received its average accounts receivable in a given period. A high number means that the firm can effectively collect the receivables and the customers pay their debts frequently, indicating a large bargaining power towards the customers. It is the fraction of the sales and the *average* accounts receivable of the period examined.

Trade payables turnover

With the help of trade payables turnover, the **payment practices of the firm towards its suppliers** is assessed. It shows how many times in the assessed period the firm has paid its debts to the suppliers. The firm has effective policies if the trade receivables turnover is large and the trade payables turnover is small. This combination indicates that the customers of the firm pay their debts rapidly and the money is not transferred to the suppliers right away. In this case the firm has significant bargaining power towards the participant firms in both the input and output markets. Accounts payable turnover is the ratio of the cost of goods sold (COGS) and the *average* accounts payable of the period.

Other industry-specific indicators

These are special indicators with **industry-specific meaning** (e.g. average revenue per customer or revenue per square meter in the retail sector).

3.5. Leverage indicators

With the help of the following leverage ratios (also called debt ratios), the firm's reliance on debt financing can be assessed.

Leverage indicators:

- Liabilities-to-equity ratio
- Interest coverage

Liabilities-to-equity ratio

Liabilities-equity ratio indicates **the proportion of debt financing and equity financing**. It shows whether the firm is indebted. It is calculated as the total liabilities (both short- and long-term) divided by the total equity.

Interest coverage

This ratio shows the **ease with which the firm can meet its interest payments**: the euro amount of Earnings Before Interest and Tax (EBIT) available for each euro of required interest payment. It indicates the degree of risk associated with the firm's debt policy.

3.6. Liquidity indicators

Liquidity indicator types:

- Current ratio
- Quick ratio
- Cash ratio
- Interval measure

Current ratio

Current ratio is definitely one the most important liquidity ratios: it shows **how many euros of current assets are available for each euro of current liabilities.** If the value is less than 1, the firm may be insolvent and not able to cover its current liabilities from the cash realised from its current assets. It is an excellent measure of short-term liquidity. Current ratio is calculated as the current assets divided by the current liabilities.

Quick ratio

Quick ratio is an improved version of the current ratio. It differentiates between assets based on their liquidity. Only the more liquid ones are included in the calculation: such as cash, marketable securities, and receivables. With the help of this indicator, one can examine **whether the firm actually has enough liquid current assets** to cover its current debts. Quick ratio is the fraction of the above mentioned liquid assets and the current liabilities.

Cash ratio

In this measurement, only the **most liquid assets** are taken into account (generally speaking: cash and marketable securities).

Interval measure

Interval measure shows **how long the firm could operate if it stopped receiving any revenues** from one day to the other.

4. Options

4.1. Definition of options

Options are contracts granting the right to the option holder (buyer) to sell or buy an underlying security at an agreed-upon price (strike price) on a specific future date. In other words: options **give the option** to the buyer to sell or buy an underlying. In the same time, options **generate an obligation** to the seller of the option.

There are two types of options: **call (buy)** and **put (sell)**. A call option offers the buyer the right, but not the obligation to buy. The seller (writer) of the option must perform on the contractual obligation to sell if the buyer decides to exercise his/her right to buy. On the contrary, a put option offers the buyer the right (but not the obligation) to sell. The seller of the option must perform on the contractual obligation to buy if the buyer decides to exercise his/her right to sell.

Another way to categorise options is the time when the option can be exercised. There are two main styles: **European** and **American**. European-style options can be exercised only at maturity, which is a specific future date. American-style options can be exercised any time between the time of purchase and maturity date.

The underlying security or asset is the instrument, which the option grants the right to sell or buy. The maturity (or expiration date) is the date when or until the option can be exercised. The **strike price** of the option is the agreed-upon price of the underlying. When buying a call option with a strike price of $50, at maturity the underlying can be bought for $50. The actual market price of the underlying at maturity does not matter.

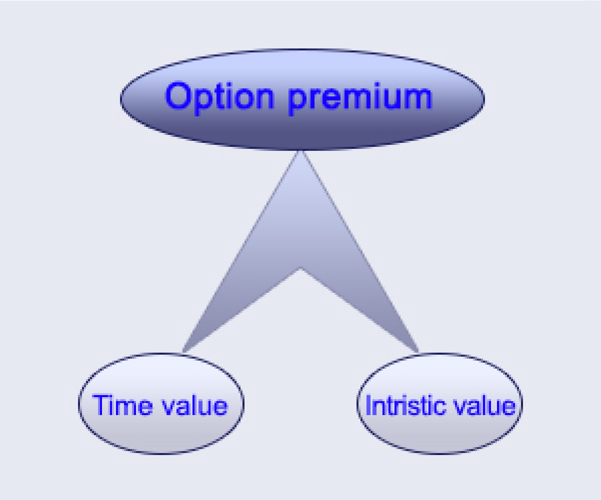
Long Call Option

However, the right to buy is only exercised if the market price of the underlying at maturity is higher than $50. If the market price is below $50, it is not worth to exercise the option, because the underlying can be bought directly on the market for less.

Options can be categorised based on their market as well. There are **exchange traded options** and **OTC (Over The Counter) options**. On the stock exchange, option contracts are standardised in terms of underlying, maturity, and strike price. Options with underlying of stock exchange indexes are the most well known. Options for the S&P and Dow Jones indexes are very common and also liquid.

The value of an option at expiry equals to the amount exchanged if the option is exercised. If the option does not worth to exercise, its value will be zero. Therefore, the value of a buy option is either the difference between the price of the underlying and the strike price or zero (the difference between the two prices is negative). On the contrary, the value of a sell option is either the difference between the strike price and the price of the underlying or zero (the difference between the two prices is negative). Before maturity, the value of the option depends on what type of option it is. **Black-Scholes** formula is used to determine European options' value with an analytic approach.

However, analytic approach cannot be used for American options, valuation is only possible with numeric methods. The most well-know numeric method is the **Binomial** model.

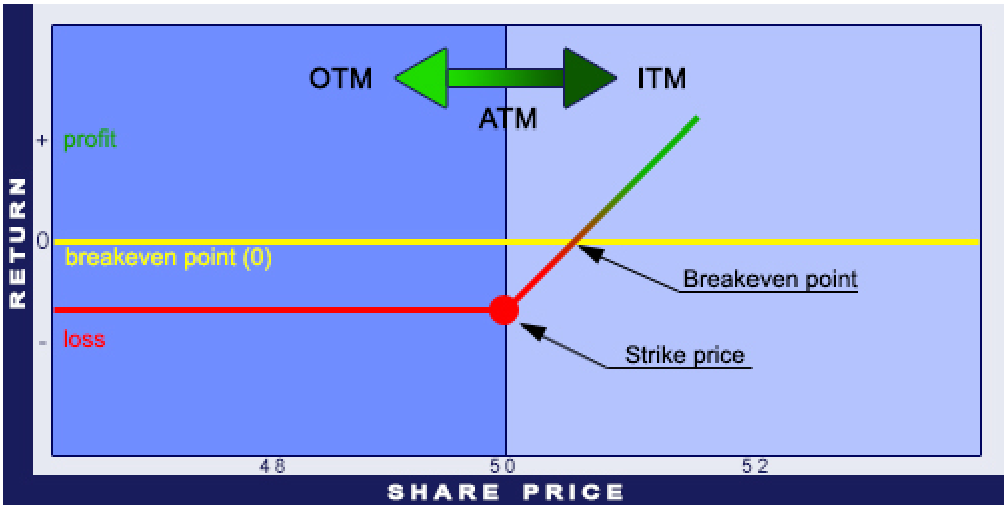


Premium of an option

The value of an option can be divided into two factors: extrinsic (time) value and intrinsic value.

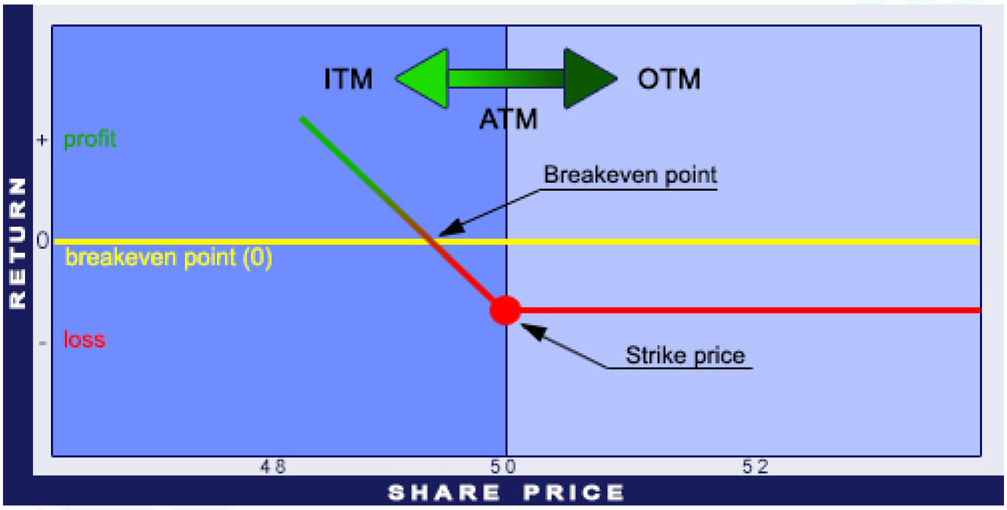
Intrinsic value is the amount an option would worth if it was exercised today (price of underlying - strike price). Time value makes up the remaining part: value of the option - time value. At maturity, the time value is zero and the value of the option equals to the intrinsic value.

In the Money (ITM) options have intrinsic values. In case of a Call option it means that the price of the underlying is higher than the strike price. For Put options it is the opposite: the price of the underlying is lower than the strike price. **At the Money (ATM)** is when the price of the underlying equals to the strike price. **Out of the Money (OTM)** options have no intrinsic values: for Call options the strike price is higher than the price of the underlying and for Put options it is the opposite: the strike price is lower than the price of the underlying. The diagram below illustrates the definition of the ITM, the ATM, and the OTM cases.



ITM, ATM, and OTM for Call options

For Put options, ITM, ATM, and OTM can be visualised the following way:



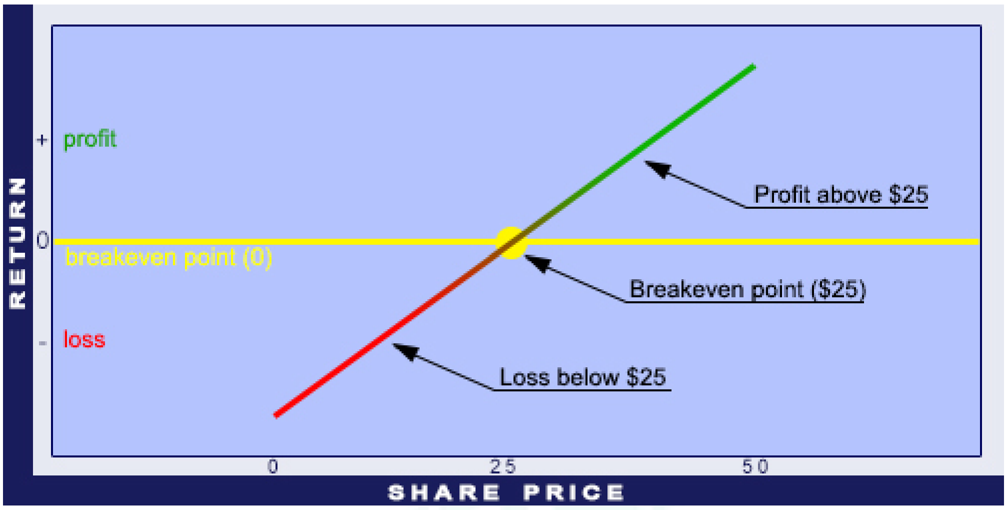
ITM, ATM, and OTM for Put options

Implied Volatility is an important factor for pricing options. The Black-Scholes option pricing model relies on this value as well. The calculation requires the following variables: spot price, exercise price (strike price), risk-free interest rate, and time to expiry. Most NASDAQ shares (especially the technological securities) have larger volatility than financial instruments traded on other exchanges, such as the NYSE. This is directly observable from the price of the options.

4.2. Risk graph

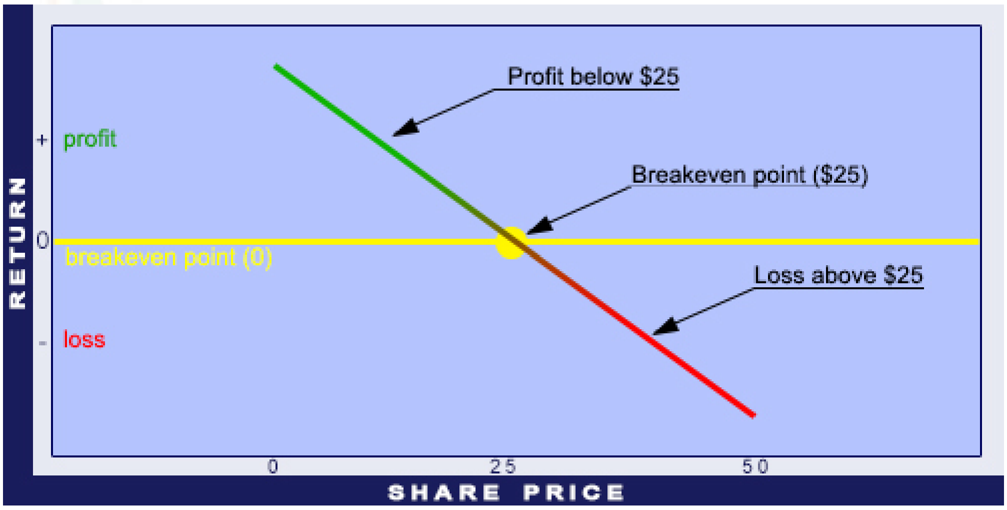
To understand deals on the stock exchange, one has to understand **risk graphs** and their use. Generally, Y axis indicates the price and X axis indicates the time. This is not the case for risk graphs. Here are some examples to illustrate how they work.

Open a long position on a share for $25. The X axis on the risk graph shows the price of the share and the Y axis shows profit/loss values. The risk graph of the share purchase is a straight line with a 45 degrees angle from the horizontal line.



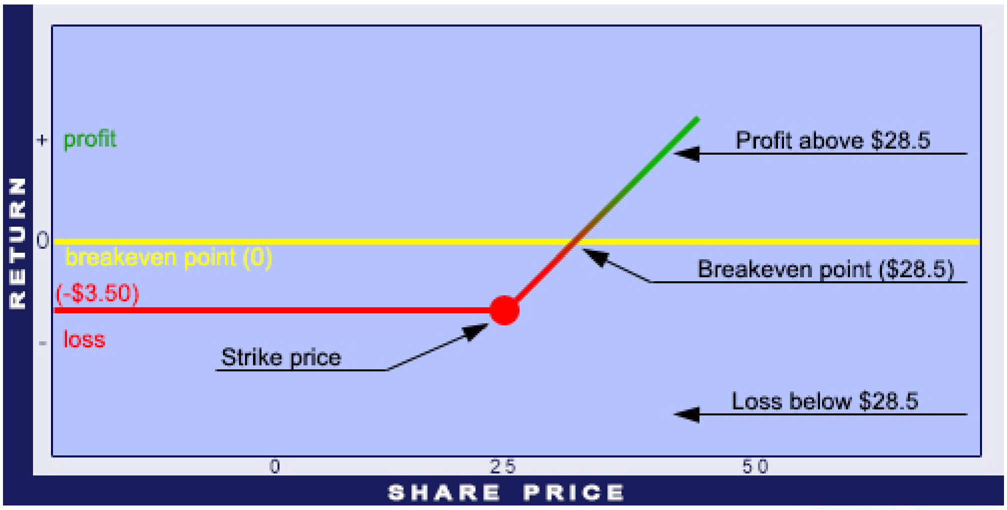
Underlying long position

Open a short position on a share for $25. Short means to sell something without actually owning the item. Opening short positions is allowed only on certain markets, for example in the US. In the UK it is not permitted. It is crucial to know that there are no limits on losses when opening short positions. Hence, the risk graph is going to look like the illustration below.



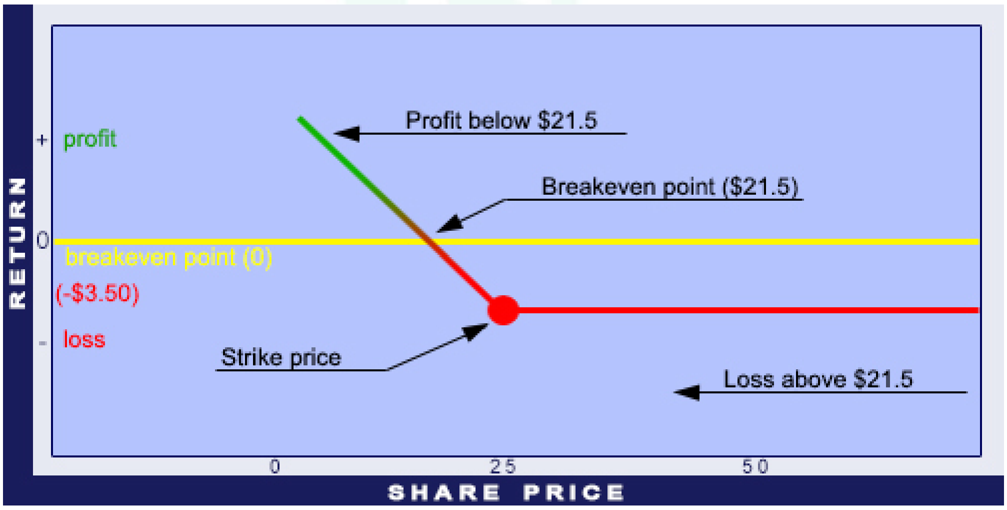
Underlying short position

Let's see the risk graph of a Call option. Call option gives the right, but not the obligation, to buy the underlying at a specified price before the maturity date. Buy a Call option with the strike price of $25 and the maturity of December. The option premium is $3.50.



Long Call position

Now let's see the same for a Put option. Put option gives the right, but not the obligation, to sell the underlying at a specified price before the maturity date. Buy a Put option with the strike price of $25 and the maturity of December. The option premium is $3.50.



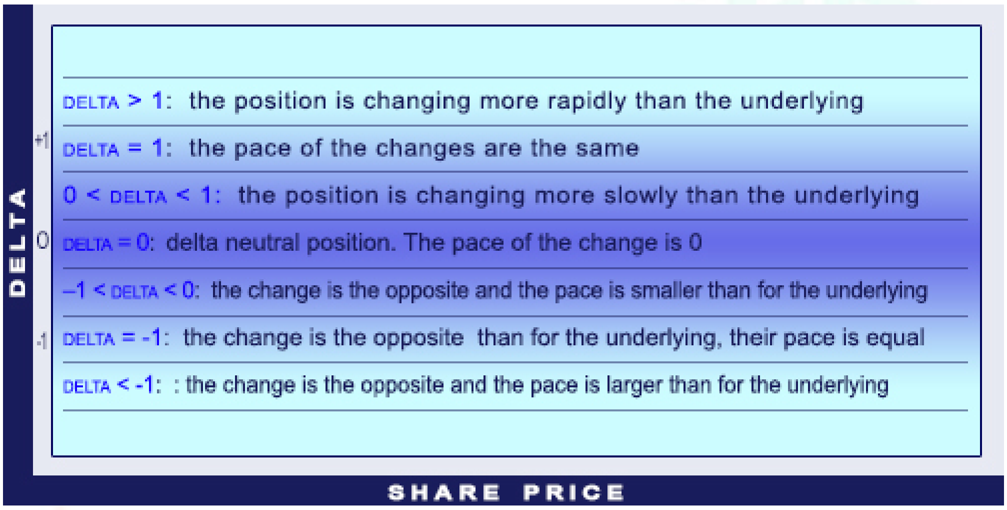
Long Put position

4.3. The Option Greeks

Delta

Delta is derived from the Black-Scholes model. It measures the effect of a $1 change in the price of the underlying on the option premium.

Thus, delta shows by how many units the premium is going to be increase if there is a one unit increase in the price of the underlying. Delta is also called hedge ratio, because it shows what percentage of the face value of the option should be purchased/sold to offset the price changes of the underlying (directional risk). The absolute value of the delta is expressed between 0 and 1. The delta of a deep OTM option is close to 0, while a deep ITM option's delta is close to 1. Close to the strike price and to the maturity, the delta is changing quickly along with the premium. One can observe that option premium does not move linearly with the price change of the underlying. Moving from OTM towards ATM makes the premium's growth accelerated. After passing the ATM point, this growth will show down. As a conclusion: when speculating with price increases, instead of buying a deep ITM option, one should buy the underlying product directly. In this case, buying a slightly OTM option can grant us higher return.



Explanation of delta

Calls have positive delta, because they are like long futures and the return is derived from the increase of the future's price.

Puts have negative delta, because they are like short futures and the return is derived from the decrease of the future's price.

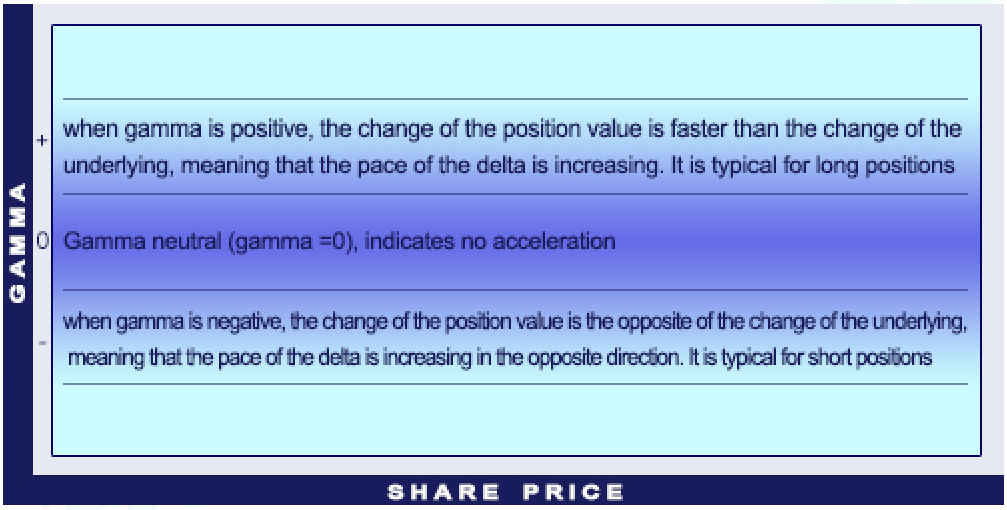
As an example: if call option has a delta of +0.5, then a $0.1 increase in the underlying's price (if all other variables hold constant) would result a $0.05 increase in the option's value. On the contrary, a $0.1 decrease would make the value of the option fall by $0.05.

Delta can be interpreted as the probability of an OTM option to become ITM option. An investor is delta neutral when selling two call options along with a future. Then the value of delta is 0. This means that selling 2 calls offsets the price movements of the future in any direction (not considering the other variables). Delta neutral situation can be achieved with two options as well.

Gamma

Gamma is derived from the Black-Scholes model. It measures the effect of a $1 change in the price of the underlying on the option delta.

In absolute value terms, ATM options’ gamma is the largest. Deep ITM and OTM positions’ gamma is close to 0. Gamma is differently observed from long and short position owners. Option buyers want positions with high gamma. When the price of the underlying changes in the beneficial direction, the delta will increase more (than for positions with smaller gammas). Thus, the premium will increase more as well. For an unbeneficial price change the delta will decrease more and the premium will lose less of its value. A high gamma intensifies the impact of positive changes and lightens the impact of negative changes. Option sellers need exactly the opposite to happen. The gamma effect is the difference between an option and a future (for which the gamma is 0). Gamma is especially important when examining the sensitivity of a delta hedge. When gamma is high, delta changes quickly and even a small change in spot price can result the position to be over- or underfunded. The closer the expiry date, the bigger the gamma’s role and the more sensitive the delta hedge.

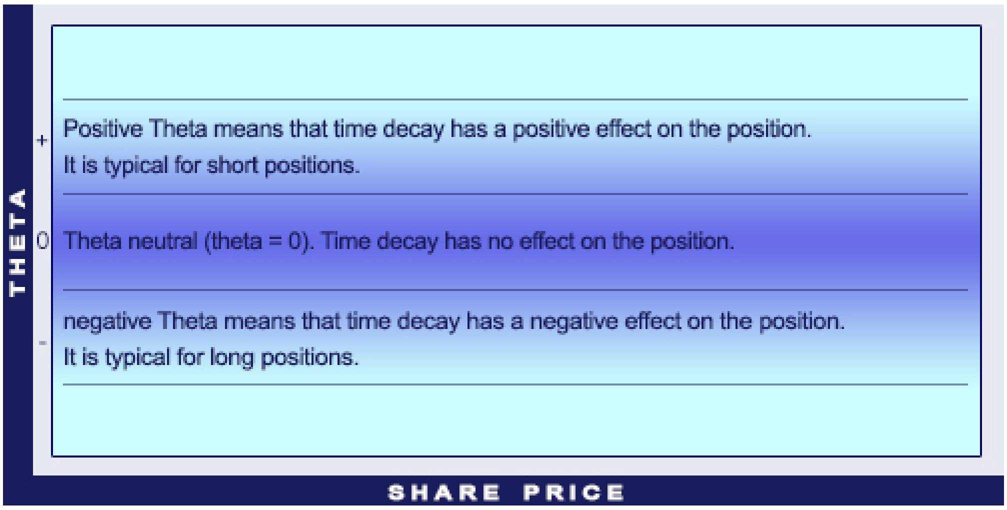


Explanation of gamma

Theta

Theta is derived from the Black-Scholes model. It measures the effect of a one-unit change in the time on the option premium.

Theta quantifies the impact of the time decay. It shows how much the option’s premium decreases for a one-unit decrease (usually one day) of the remaining time until maturity. The closer the maturity, the faster Theta grows and the more value long positions lose. The acceleration of the time decay is more intense for ATM positions than for ITM and OTM positions. The more ATM the position, the larger the theta (if the expiration is the same). High theta is beneficial for the sellers of the options (short position). There is a special exchange between theta and gamma. It is true for both cases that the closer the expiration and the more ATM the position the higher the value of the Theta. However, high gamma is beneficial for long options, but because of the increased time decay, large theta is beneficial for short options.

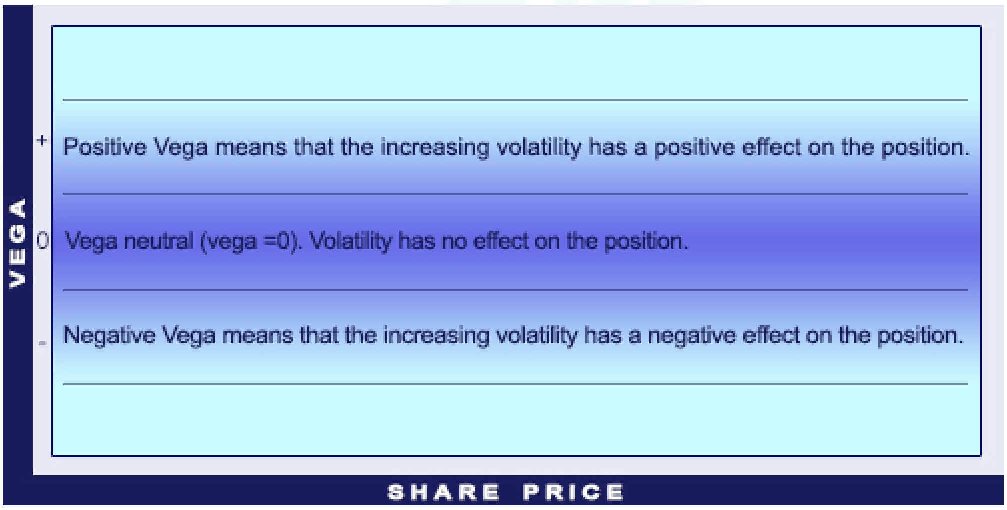


Explanation of theta

Vega

Theta is derived from the Black-Scholes model. It measures the effect of a one-unit change in the volatility on the option premium.

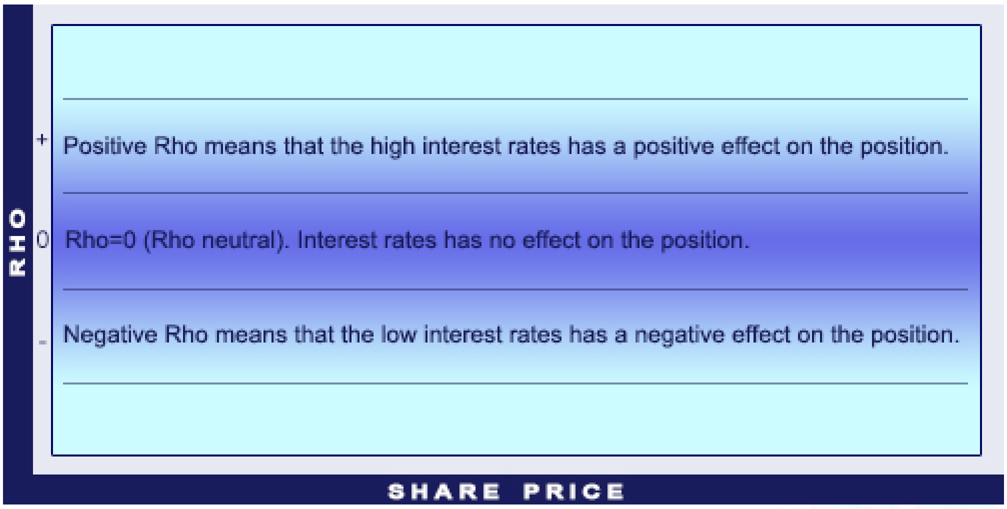
The farther the expiration and the more ATM the position, the higher the vega.



Explanation of vega

Rho

Rho shows the impact of interest rates on the value of the option.



Explanation of Rho

Summary

GREEK	INFLUENCE
Delta	Correlation between the price of the underlying and the price of the option (velocity)
Gamma	Correlation between the price of the underlying and the delta of the option (acceleration)
Theta	Correlation between time to maturity and option value (time decay)
Vega	Correlation between (historical) volatility and option value
Rho	Correlation between the interest rate and option value

GREEKS' SIGNS IN DIFFERENT POSITIONS				
Position	**Delta**	**Gamma**	**Vega**	**Theta**
Long Call	+	+	+	–
Short Call	–	–	–	+
Long Put	–	+	+	–
Short Put	+	–	–	+

4.4. Option orders

For a beginner option trader differentiation between the different order types and their several parameters may be difficult and confusing. Trading of shares is much easier as there are only two order types: sell or buy. In options trading there are four types of orders: Buy to Open, Buy to Close, Sell to Open, and Sell to Close.

Every order has several ways to customise. Orders with the wrong parameters can result the biggest lost.

Buy to Open

Buy to Open is the most common order in options trading. The term means to open a position via buying an option contract. Thus, having a buy or long position.

Example:
John wants to buy a Call option with the following parameters: expiry date: January; underlying: Apple share; strike price: $40. John executes the following: *Buy to Open Jan40Calls*.

Sell to Close

Sell to Close is the second most popular option order. The term means to close a position via selling an option contract. Thus, selling one of the previously acquired option contract.

Example:
John wants to sell his Call option with the following parameters: expiry date: January; underlying: Apple share; strike price: $40. John executes the following: *Sell to Close Jan40Calls*.

Sell to Open

Sell to Open is a less frequent type of option order. The term means to open a position via selling an option contract. In other words, having a short position. That means option writing.

Example:
John wants to create a Short Call option strategy with the following parameters: expiry date: January; underlying: Apple share; strike price: $40. John executes the following: *Sell to Open Jan40Calls*.

Buy to Close

Buy to Close is an order to close a Sell to Open position. It means to close a position by buying an option. Thus, repurchase an option contract that has been sold previously.

Example:
John wants to close his Short Call option with the following parameters: expiry date: January; underlying: Apple share; strike price: $40. John executes the following: *Buy to Close Jan40Calls*.

Combination orders

Option strategies are usually complicated and require the combination of the above mentioned orders. For instance, a Bull Call Spread strategy consists of the following actions: buying an ATM Call option and selling an OTM Call option. Thus, to open the Bull Call Spread strategy one must Buy to Open an ATM Call option and then Sell to Open an OTM Call option. When closing the position, the investor must Sell to Close the ATM Call position and then Buy to Close the OTM Call option.

There are two option order strategies which are executing both the opening and the closing of the position. These orders are called *One-Trigger-Other* (OTO) and *One-Cancel-Other* OCO.

One-Trigger-Other (OTO)
The OTO order activates another option order, when the initial order is executed. It is especially useful for Covered Call option positions, when the owner of the position wants to sell the call option right after the purchase of the share.

One-Cancel-Other (OCO)
OCO is less well-known than OTO, yet it is an extraordinarily useful order type. When one order has been fulfilled, OCO executes the cancellation of the other order. The option trader protects his position from losses (with a stop order) and ensures gains (with a limit order). This is important when one wants to define the *Stop Loss* and the *Profit Taking* levels.

Execution orders

A beginner option trader must know at least two type of orders out of the many. These two are the Market Order and the Limit Order.

Market Order
To execute an order at the market price, one must choose *Market Order*. In this case the investor asks the broker to execute the order at the first available price, without any consideration to the price levels. This order type can be used successfully only on the liquid option market and with really tight spreads, where prices do not change rapidly. On illiquid markets the option order would be executed at a really high price. The biggest disadvantage of the Market Order is that one can never be sure at what price the order will be executed. This may be a critical factor for money management. The actual execution price may be higher than the price seen in the books, thus one must count with some reserves. This is why brokerages ask for margin for Market Orders.

Limit Order
To execute an order a chosen price, one must choose *Limit Order*. In this case the investor tells the broker to open a position, but it cannot go above the chosen price. In this order, the investor will know how much it will cost maximally; therefore it requires no extra margin. The biggest disadvantage of Limit Orders is that the prices can move downwards as well, in which case the order will not be fulfilled. When the order has not been executed, one must look at the direction of the movement of the prices from the limit and modify it.

Time in force

It may not be enough to tell the broker whether to execute an order at market or at limit price, the validity period should be specified as well. The most popular time-based orders are the following: *Good Till Cancelled* (GTC), *Day Order* (DO), *Fill-Or-Kill* (FOK), *All-Or-None* (AON), and *Market-On-Close* (MOC).

Good Till Cancelled (GTC)
GTC is an option order which is valid until manually cancelled. So it does not expire at the end of the day. GTC orders are useful when defining *Stop Loss* or *Profit Taking* levels, but unsure when the price will reach those levels. Most option traders have this as a default setting.

Day Order (DO)
DO loses its validity as soon as the market closes and the order is canceled if it was not executed during trading hours. This is advantageous when opening more option positions, but do not want any pending orders at the end of the day.

Fill-Or-Kill (FOK)
FOK is either forthwith executed or forthwith cancelled. Therefore, FOK is only valid for a moment. If it is not executed immediately, it expires. It is beneficial for traders with portfolio management strategies requiring full orders to be executed.

Immediate-Or-Cancel (IOC)
IOC is somewhat similar to FOK. The difference is that IOC allows the position to be partially executed, then cancels the rest that could not be fulfilled. Therefore the position can fully or partially fulfilled after release. If the order cannot even be partially executed, it becomes canceled.

All-Or-None (AON)
AON can only be executed fully, but not necessarily immediately. AON differs from FOK and IOC, because the order does not expire if not executed immediately. It can be used together with DO and GTC as well.

Market-On-Close (MOC)
In case of an MOC, the current day's closing price will be implemented. This order may be handy for investors believing that the closing price is the most beneficial on the given day. It is also beneficial for the option writer who wished to maximise his/her profit by letting the time value grow until the day closing.

Exit orders

After deciding the position type to be opened, the investor must specify an exit level. This level is either called *Stop Loss* or *Profit Taking* level. To eliminate emotions from the trading, option traders can give Stop Orders, Contingent Orders, or Trailing Stop Orders.

Stop Orders
A Stop Order is when determining a so-called *Stop Price*. For long positions it indicates the level above which the order must be executed. In case of short positions, it is the level under which the order must be executed. It is important to secure an already achieved gain. Example: an investor has bought shares for $100 a year ago. The current price is $250, but there is a chance that an uncertainty on the stock market can result a fall in the prices. The investor wants to make sure to gain at least $100 on the position and gives a Stop Order with a stop price of $200. When the market price reaches $200, the order will be executed and the investor will realise the secured profit. The only disadvantage is that the order is only executed when reaching the stop price. Sometimes market gaps can prevent Stop Orders from being executed, resulting massive losses.
There are two types of Stop Orders: Limit Stop and Market Stop.

Limit Stop
Limit Stop Order is when the investor determines a Stop and a less beneficial Limit Price to prevent uncertainties in the Stop Orders. When the underlying share's price falls so quickly that after passing the $200 price level, the next trade is only at $170, the Stop Order can be executed only at that low price. To prevent that, the investor can determine a limit price as well, for instance at $190. Under that price there will be no sale happening.

Market Stop
Market Stop Order activates the market order when the price level is at the stop level. Market Stop Order grants to close the position when the price level reaches the stop level. Its disadvantage is that the execution price can be much lower than the stop order price.

Contingent Orders
Contingent Orders are a combination of other order types. It consist of a parent order (a market price, a limit, or a stop order) and one or two supplementary orders. The supplementary orders are only triggered once the parent order is fulfilled. The Contingent Order can be Stop Order and/or Limit Order as well. Such condition for an option position can be to close the position only if the price of the share has dropped below or raised above a certain level. When the investor creates a stop and limit order in the same time, there is a possibility to use them as an OCO.

Trailing Stop Order
Trailing Stop order is a special type of Stop Orders. A stop order that can be set at a defined percentage away from a security's current market price. This can mean only in one direction, depending on whether it is a buy or sell position. For a stop buy, the stop price will be adjusted downwards when prices drop. When prices rise, the stop price will remain unchanged and the order will be automatically exercised once the price reached the actual stop price. For stop sell it is the contrary: When prices increase, the stop price will be adjusted upwards. When prices fall, the stop price will remain unchanged and the order will be automatically exercised once the price fell to the level of the actual stop price. The option trader can tell the broker to close the position when the option price is 10% lower than the highest price. The broker will automatically update the highest price of the option.

4.5. The risk of options

Trading with options is risky. But what are the risks associated with option trading? Which option strategies are risky and which aren't? Is it possible to lose all the invested capital with option trading? The following lesson is looking for the answer to all these questions.

What are the risks of option trading?

What does risk really mean on the stock market? The risk is the possibility to lose all the invested capital. Simply put, it is the probability to lose money. Trading options is riskier than trading shares, because of its leverage effect. The incorrect execution of option strategies may result significant losses.

The official description of the risk in option trading

The Characteristics and Risk of Standardized Options document, published by the Chicago Board of Exchange (CBOE), is a good start to define options' risk. Chapter X. is dedicated to this topic. The risks are assessed based on the buyers and the writers of the options and other risk factors are also evaluated. In the following paragraphs, the stock market options are analysed.

Risks of option trading from the buyer's perspective

1. The money invested in options can be lost quickly, the premium of the bought option may be close to 0. Example: There are three investors: A, B, and C. All of them own $5,000, and hope that the price of XYZ share will rise. The current market price of the share is $50. *Investor A* invests his $5,000 in 100 pieces of XYZ shares. *Investor B* invests $500 in Call options with the strike price of $50 (the premium is $5, thus he is entitled to buy 100 shares). He invests the other $4,500 in treasury bonds, which is a relatively risk-free investment. (The expiration of Call option in the example is 6 months. The annual return of the treasury bond is 3.25%, so the return after 6 month is $73.) *Investor C* invests all his money in Call options (10 contracts with $50 strike price with $5 premium). The following table evaluates the possible gains and losses 6 months later with different market prices.

The market price of XYZ share at the option's expiration	Investor A		Investor B		Investor C	
	Profit / loss	Return	Profit / loss	Return	Profit / loss	Return
62	+ 1200	+ 24%	+ 773	+ 15.7%	+ 7000	+ 140%
58	+ 800	+ 16%	+ 373	+ 7.5%	+ 3000	+ 60%
54	+ 400	+ 8%	– 27	+ 0.5%	– 1000	– 20%
50	0	0	– 427	– 8.5%	– 5000	– 100%
46	– 400	– 8%	– 427	– 8.5%	– 5000	– 100%
42	– 800	– 16%	– 427	– 8.5%	– 5000	– 100%
38	– 1200	– 24%	– 427	-8.5%	– 5000	– 100%

The table clearly shows that investing in options can generate both big profits and losses. *Investor C* - who invested all his money in options - has achieved the largest return when XYZ's market price was $62. However, his position was already making losses when the market price at expiry was $54 (because the time value of the option run out). If the market price didn't change at all, the whole investment would have been lost and the option would have lost its value and expired.

2. The higher the OTM level of the option, and the closer the option to expiration, the bigger the probability that the capital will be lost and the level of risk increases. The more OTM an option the less probable it will become ITM. With the approaching expiry date, the number of days to change to ITM decreases and the risks further increase.

3. European options cannot be executed before expiration date. The only way to realise profits before expiry is to sell them.

4. Certain options have risks at execution. For example if the brokerage changes the costs of the execution, or when there aren't enough funds available on the account for the automatic execution of a Call option. In this case the option will expire worthless and lose its value.

5. Courts or other authorities (e.g. *Options Clearing Corporation* (OCC)) can introduce enforcement limitations which prevent to realise profits.

Risks of option trading from the seller's perspective

1. The written options can be executed any time before expiration. Although American options can be executed before expiration, in reality early assignment only happens with ITM options shortly before expiration. When the buyer executes the options, the seller must deliver the underlying security (Call option) or must buy the underlying security (Put option).
2. Covered Call traders give up the right for further profits as soon as the share price rises above the strike price of the option. However, they accept the risk that the underlying's price may decrease. Example: The market price of XYZ is $50 and the investor writes a Call option with the strike price of $50 and the premium of $4. As the expiration is getting closer, the price of the share increases to $58 and the buyer executes the option. The profit - apart from the dividends - is the premium of the Call option. Without writing the option, the investor could have gained money from the $8 increase in the market price. On the other hand, if the underlying share's price decreases below the strike price and the option is not executed, and it will result losses.
3. The writers of uncovered Call options (Naked options) agree to face unlimited losses if the underlying's price increases above the strike price. When the Call option is executed, the writer must sell the shares for the price agreed in the contract. Thus, a sudden price increase can result significant losses for the writer of a Naked Call option.
4. The writers of uncovered Put options agree to face unlimited losses if the underlying's price decreases below the strike price. When the Put option is executed, the writer must purchase the shares for the price agreed in the contract. Thus, a sudden price decrease can result significant losses for the writer of a Naked Put option.
5. The writer of a naked option undertakes the coverage risk if his position generates losses. Brokers grant liquidity to hedge such risks.
6. Writers of Call options can lose more money on the same price increase than on a short position of the share. If the investor sells 100 XYZ shares for $5,000, then with a $12 price increase (from $50 to $62), the loss will be $1,200. Whereas, when writing 10 Naked Call options with the strike price of $50, the same price increase would result a loss of $7,000.
7. The writer of the Naked Call must deliver the shares for the strike price when the option is executed.
8. Options can be executed after the market closes.
9. Writers of options have the obligation even when the market is unavailable, thus they may not be able to close their positions.
10. The price of the underlying can increase/decrease so quickly that the options get automatically executed.

Other risk factors

1. The complexity of some option strategies are a significant risk in itself. This is especially true for complex portfolios based on selling and buying options. Writers of Straddle options must face unlimited risk.
2. The option markets and the option contracts are continuously changing. The conditions and validities are not constant.
3. The option market has the right to suspend the trading of any options, preventing to realise profits.
4. Incorrect execution of options may occur.
5. When an option brokerage goes out of business, the investors can be harmed.
6. International options bring special risks because of the difference in the time zones.
7. A change in the options' implied volatility can result losses even when the share prices move in the expected direction.

The factors of the options'micro risks

In *The Characteristics and Risk of Standardized Options* the risk exposure of option buyers and sellers have been summarised comprehensively. Now the risks are going to be examined on the micro level.
1. Uncovered option positions come with unlimited risk.
2. Options can expire worthless. When this happens, the invested money is going to be lost. This is true for 90% of the written options.
3. The leverage effect of options can be useful and dangerous in the same time.
4. Obligation can be highly risky.
5. Conditions of specific option contracts can be changed anytime by the option market or the option brokerage, within legal limitations.

All the factors above are significant risks on the invested capital, thus it is inevitable to be aware of all of them.

The factors of the options' macro risks

There are three macro risk factors on the stock market. They are not necessarily true for option trading exclusively. These are the *primary risk (market risk)*, *secondary risk (sector risk)* and *idiosyncratic risk (individual stock risk)*. Options are derived from other securities, thus their risk depends on the underlying product's risk. That's why it is important to be aware of these risks as well.

Primary risk (market risk)
Primary or market risk is when the market moves in the opposite direction than expected. If an investor owns a long Call, then the primary risk is that the market prices fall and the Call option becomes OTM. The more shares are bought (the more diverse the portfolio) the bigger the probability that the portfolio will move together with the market. DOW index is a good example, because it consists of more than 30 shares. When investing in all 30 shares, one gets exactly the movement of the DOW index. This is a significant option risk, because the investor is in an overall long Call position.

Secondary risk (sector risk)
Secondary or sector risk is when the sector moves in the opposite direction than expected. It can happen that the shares of specific sectors do not follow the movement of the market. This may result a sector-wide decrease in the market prices. This risk is relevant to the trader if he used a bullish strategy on shares from the same sector.

Idiosyncratic risk (individual stock risk)
Idiosyncratic or individual stock risk is when one invests all his money in the shares of one firm exclusively. It is because any news related to the company can negatively influence the movement of the share price. When investing in only the shares of XYZ, one faces the overall risk of the firm (also the default risk). Individual stock risk happens in option trading when all capital is invested in options with the same underlying share.

There is no chance to hedge all risks mentioned above. When hedging the primary risk by buying only a few shares, the secondary and the individual stock risks are intensified. When individual stock risk is hedged by buying shares of firms in the same sector, the secondary risk is increased.

Delta risk

Delta risk is the one affecting option trading the most. The option strategy does not count, the underlying must behave according to the chosen option strategy to generate profit. For example, applying a Bull Call Spread strategy on the shares of XYZ requires the share prices to increase. If the forecasts are wrong, the trader will lose money. Delta risk can be hedged by a delta neutral strategy.

Other risks

There are other risks apart from the delta risk. These are the gamma, rho, vega or theta risks. They can be hedged by spread trades.

Main lessons

Option trading is risky and one may lose the whole invested capital. However, the risks can be decreased by recognising and hedging the biggest risk factors. Choose an option strategy which is profitable from more directions and be careful with the leverage nature of options. An option trader must be aware of all risks he is facing. After all, option trading is not necessarily riskier than stock trading.

Stock trading can be riskier than option trading

Option trading is risky, but is safer than stock trading from the following aspects. The term *risk* indicates the probability of losing the invested capital. Below it is shown which stock trading positions can result higher risks than option trading.

Share and leverage option

Shouldn't options be riskier than shares because of their leverage nature? Yes, when leverage is handled incorrectly. No, if it is managed correctly. Option contracts enable the trader to invest and profit from market price movements for the fraction of the actual share prices. It means that with option trading a smaller part of the capital is risked with the same underlying; decreasing the overall risk. Such position can be the combination of a long Call and a short Put, which is also called a synthetic stock.

Example:
An Apple share is traded at $43.57 and a Call option with the strike price of $43 and maturity of January has the premium of $1.63. One can buy 100 Apple shares for $4,357 or buy 100 Call options for $163. Then the investor profits from an increase in the share price. The maximum loss in the case of the shares is $4,357, whereas the maximum loss is only $163 for option trading. The biggest mistake is when one buys Call options for $4,357. This is a surprisingly common mistake.

Short selling shares

The only way to profit from a decrease in the share price is to short the position. To short something means to sell without owning the product. When the price of the share increases suddenly, investors may lose all their money and may even have issues with the margins. In case of option trading, buying a Put option will benefit from the same instrument's price decrease. However, losses are limited by the option premium.

Example:
Apple shares' price is $43.54. The premium of a Put option is $0.70 (expiration in January, strike price is $43). The stock trader shorts 100 shares and the option trader buys 1 contract for $70. If the price of the Apple share increases to $50, the stock trader will lose $646, whereas the option trader will only lose $70.

Shares generate profit only in one direction

Trading shares generates profit only if the share price increases (Long position) or decreases (Short position). But both directions cannot be used to earn profit. However, there are option strategies which generate profits independently from the direction of the price changes of the underlying. Investors can also profit from sideway prices. The more ways a strategy can profit, the bigger the chance to earn profit and the smaller the risk. The most popular option strategy which generates profit in both price increase and decrease is the Long Straddle.

There is no hedge in stock trading

Hedging is when high risk positions are mitigated by a lower risk position to reduce risk exposure. When investing in shares, the only possibility is to diversify the portfolio. In option trading high risk positions can be prevented by other options or shares.

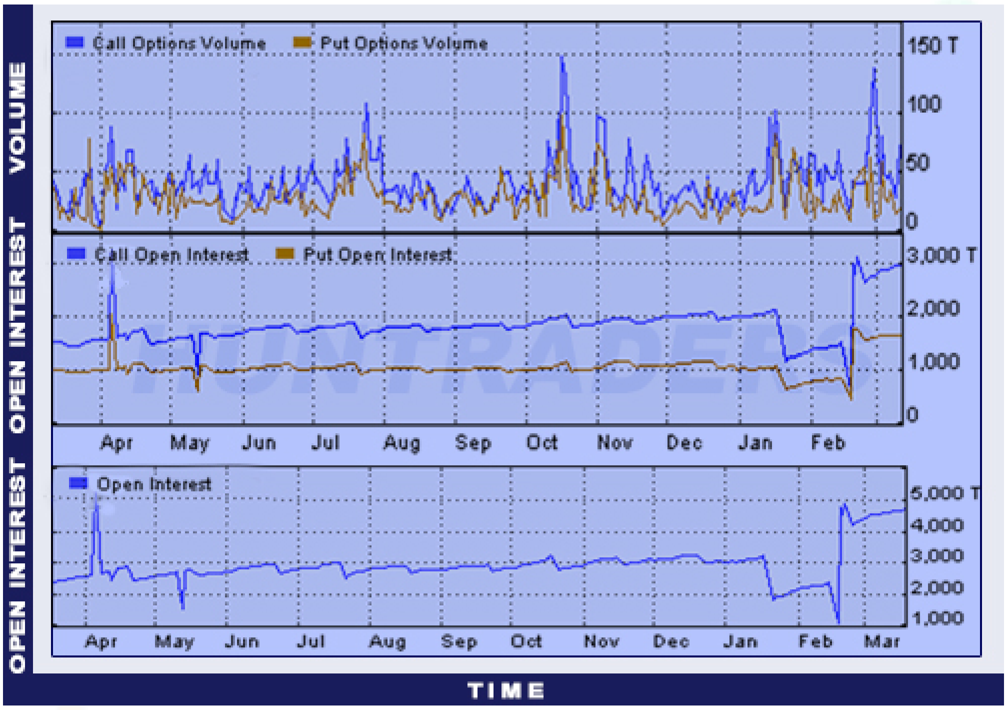
Main lessons

Options are more complex products than shares. In option trading there are more security possibilities built in and only the wrong implementation of the leverage factor can make options riskier than shares. Neglecting these rules makes option trading highly risky. However, one can make option trading safer by keeping the above mentioned advices in mind.

4.6. Option Volume and Open Interest

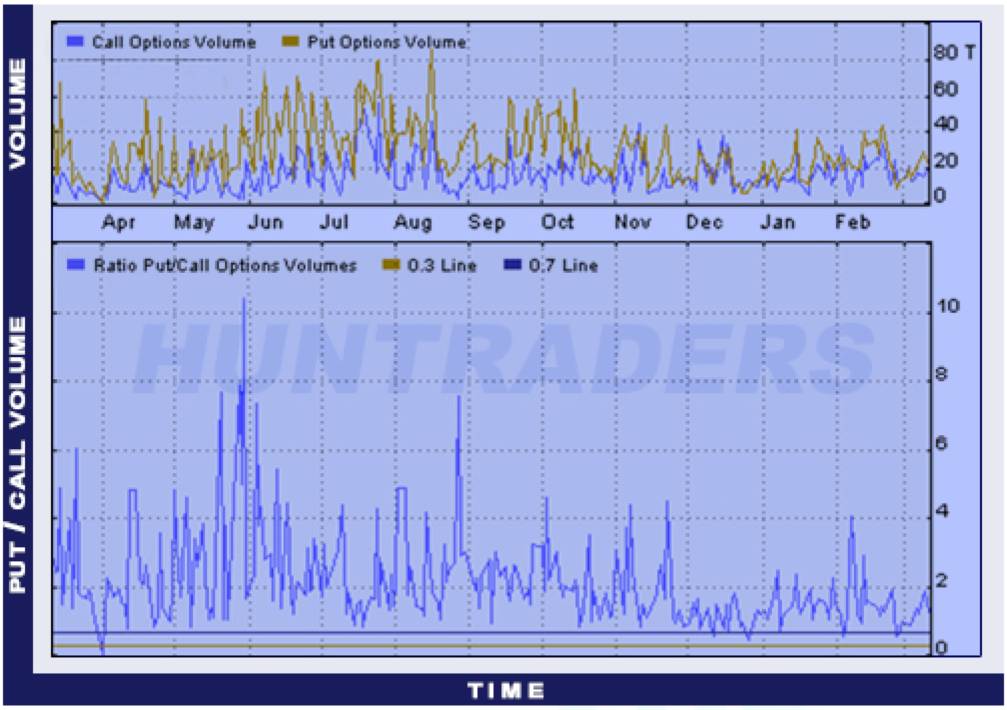
Volume means the contracts traded on a single day. **Open Interest** is the number of open positions related to a given option. An option position is open, if it has been bought and hasn't been sold or the option right hasn't been executed. The two are not necessarily the same data, although it might be confusing first. If the Volume is low, the Open Interest can be still high. The higher the value of the Open Interest, the bigger the chance to close the position and the higher the option's liquidity. Another way to measure liquidity is to observe the value of the *Bid/Ask spread*. Bid is the price for which traders can sell an option, Ask is the price for which traders can buy an option. If the difference is high between the two then it has a negative effect on the profitability.

As an example, the Open Interest and Option Volume indicators of MSFT are shown for both Put and Call options. It is clear that the activity of Call options exceeds the activity of Put options. It means that investors expect the prices to increase.



Open Interest and Volume of Call and Put options on an MSFT share

The next chart shows DJX index. The increasing Put/Call ratio indicates a growing bearish market mood, when the investors buy Put options to hedge.



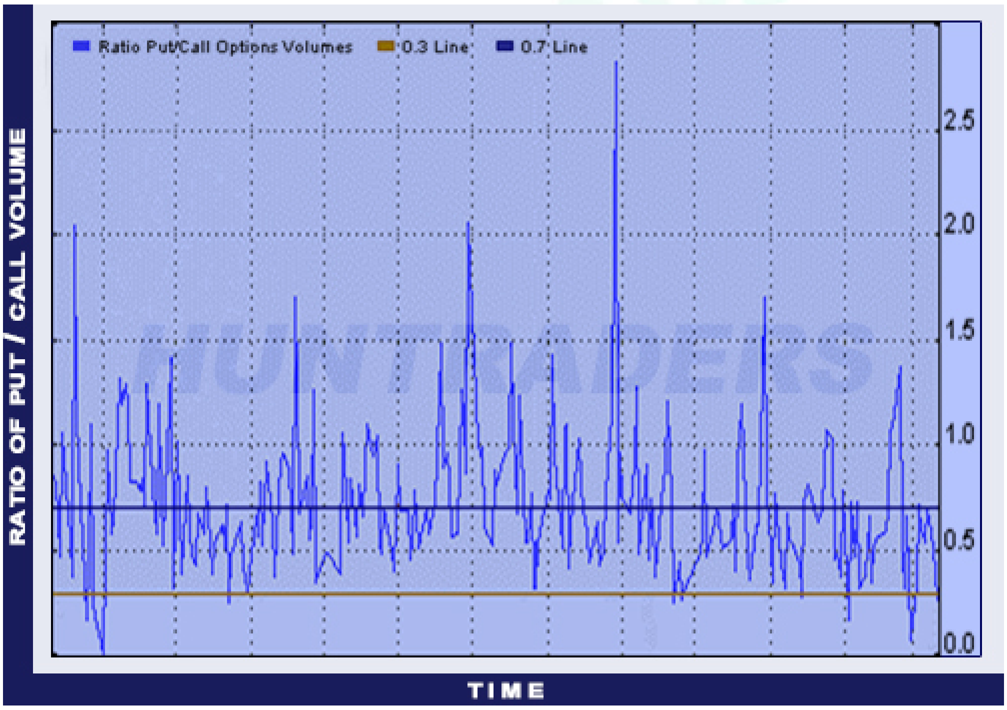
Volume of Call and Put options and the Put/Call ratio on a DJX index.

4.7. Put/Call ratio

The trading volume of put options to call options is useful for two reasons (regarding the premiums of long and short options). First, it shows how much the expectations of the market participants differ from the futures market's prices, and how they value the put premiums compared to call premiums. Second, it can predict a direction change in the futures market's prices, indicating possibilities to buy undervalued premiums or to sell overvalued premiums. **Put-Call ratio** is based on the assumption that buying intention is stronger (e.g. everyday investors rather buy than sell options). Therefore, **increasing Put-Call ratio indicates bearish, decreasing ratio indicated bullish sentiment on the market**. Then investors act completely the opposite:

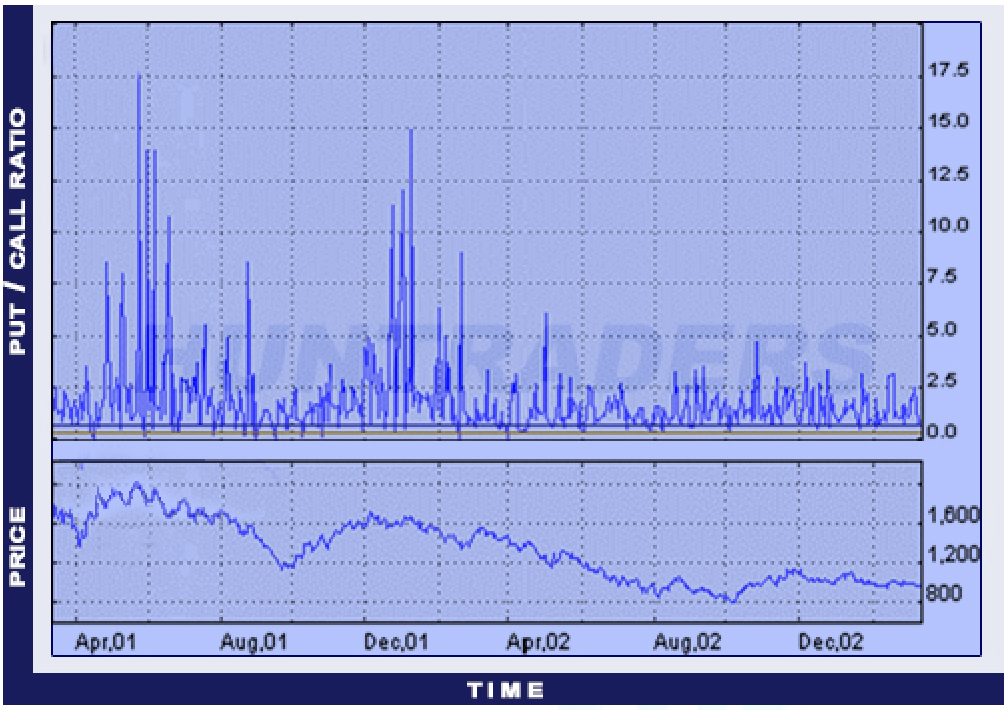
- **a high Put-Call ratio** signals them to buy call and sell put options,
- **a low Put-Call ratio** signals them to buy put and sell call options.

To determine what *high* and *low* values mean, one must analyse the Put-Call ratio's historical data. The interpretation of the Put-Call ratio is highly flexible. In option trading the best is to look at the ratio as an indicator of **possible extremities**. There is a market euphoria under 0.3 and there is a market pessimism above 0.7. The chart below shows the Put-Call ratio for MSFT. The low values indicate the expectation of increasing prices.



Put-Call ratio for MSFT shares

The Put-Call ratio for the NASDAQ-100 (NDX), on the next chart indicates an **intensifying bearish** market mood.



Put-Call ratio for NASDAQ-100 index

It is not easy to decide upon the sale or purchase of options. Apart from that, the buying intention is not always dominating the market. However, constantly monitoring the Put-Call ratio - along with other technical analysis tools - can be really useful:

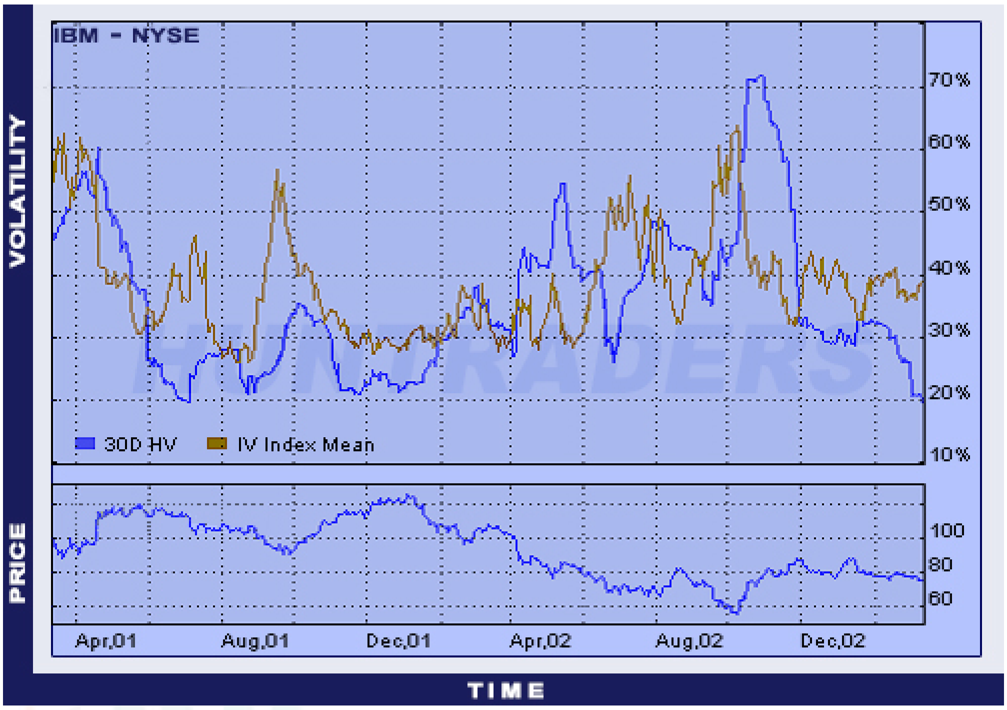
- **On a demand-oriented market** a high Put-Call ratio can signal to buy Put options or to sell Call options, before the peak of the futures' prices.
- **On a supply-oriented market** a high Put-Call ratio can signal to buy Call options or to sell Put options, before the lowest point of the futures' prices.

On demand-oriented markets, the high Put-Call ratio means that the traders release plenty of OTM Short Put, hoping they would expire worthless and lose their value. Because of the increasing bullish market mood, they avoid to sell Call options. However, after a certain point the bubble pops, because the prices of futures cannot grow indefinitely. When the relapse occurs, the writers of Put options start to hedge their positions immediately, and will find it more attractive to sell Call options. This sequence of events leads to a decrease in the value of the Put-Call ratio. The ones who have seen it coming already have Long Put and Short Call positions and profit from the thoughtlessness of other traders for two reasons. One is that the Put options were probably cheaper than Call options before the fall in futures' prices, as investors sold more of them. The other reason is that the purchase offers on Put options would increase the premium to such high level that would exceed the value of the justifiable fall in the futures' prices. Investors will try to close their positions and volatility will increase along with the purchase offers.

The opposite of this is true for the **supply-oriented markets**. The frequency of a Put-Call ratio with high value probably indicates that the investors own mainly Long Put options and try to avoid Long Call options. This situation is not generally applicable after a drawdown in futures' prices, but it motivates traders to close their positions or to buy Call positions. On supply-oriented markets low Put-Call ratios are more frequent, because apart from buying Put positions, traders also write Call options, hoping they will expire worthless and lose their value when the futures' prices fall even lower. It may more beneficial to write Put options or to buy Call options in this case. On supply-oriented markets a low Put-Call ratio suggests that investors should sell Call options and buy Put options.

4.8. The definition of volatility

Volatility is the measurement of the fluctuations in market prices. When the market price of the underlying increases/falls by 15 or 20 bps daily, the market is more volatile than in the case of 3 or 5 bps fluctuations. When the volatility of the underlying is 27%, then an underlying with the face value of $7.85 is expected to move between 7.85 ± 2.12 (2.12 = 7.85 x 0.27) this year. The probability of the price staying within this range is 68%. When volatility increases, share prices decrease. The following graph shows the historical volatility, the implied volatility, and the market price of the IBM shares.



Volatility and share price of an IBM share

4.9. Calculating volatility

Volatility is the variability of the return. This variability is measured by the standard deviation of the return with continuous interest rate payments. Volatility is usually calculated from the daily closing prices, but weekly, monthly, hourly, or even minutely data could be used. If volatility is constant, it doesn't matter what data is used, because the $\Delta T$is the variance per time unit $\sigma^2 \Delta T$, and it is possible to calculate the value of σ from that. The formulas needed to calculate volatility is the following:

$$z_t = ln(\frac{S_t}{S_{t-1}}) V = \sigma^2 \Delta t = \sum_{k=1}^{n} \frac{(z_t - z)^2}{n-1}$$

where n is the number of observations.

$$\sigma = \frac{\sqrt{V}}{\sqrt{\Delta t}}$$

The default time unit years. When using weekly data the following amendment must be made: $\Delta t = \frac{1}{52}$. If σ_{weekly} indicates the weekly volatility calculated from weekly data, then the relationship between weekly and yearly volatility is

$$\sigma = \frac{\sigma_{weekly}}{\sqrt{\frac{1}{52}}} = \sigma_{weekly} \sqrt{52}$$

When volatility is calculated from daily data, then the formula to adjust the result is:

$$\sigma = \sigma_{daily} \sqrt{252}$$

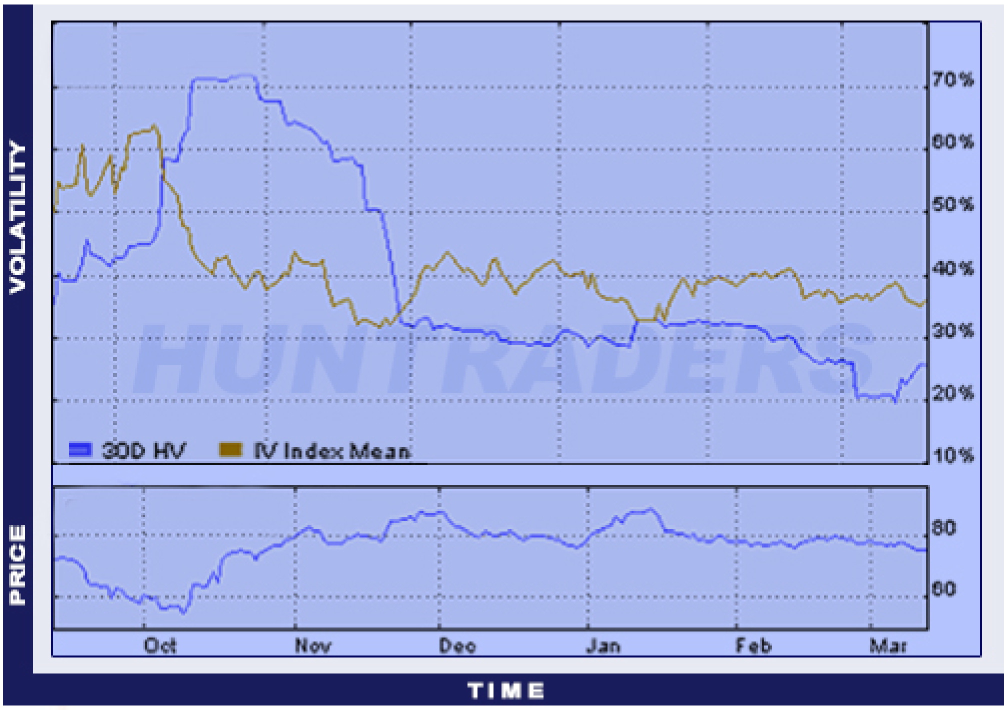
252 is used because there are approximately 252 business days in a year and statistical analyses show that market prices are more dependent on the stock exchange business days than on calendar days (on weekends, the market price changes are significantly lower than on workdays). It is important to note at this point that the *n* number of observations does not equal to how many parts one wants to divide the analysed period.

The higher the volatility the higher the fluctuation in the market prices. **Statistical volatility** is the percentage value of the price fluctuations determined by statistical methods. Traders and analysts examine the *normal distribution* of the closing market prices in a given period to determine the value of the statistical volatility. When assuming the price cannot be zero or negative, *lognormal distribution* may be a better choice, but it requires more data processing. The 22.6% volatility is simply the standard deviation of the price ratios' lognormal sequence on the 252nd business day and this is how one can get the yearly statistics. The sequence can be formed from the closing prices on 20 days when this period can realistically reflect the market price fluctuations. More or less than 20 days' data can be used as well to build the sequence, but it is highly recommended to use a 15-20 days long period as a basis to get a statistically acceptable result. When the volatility of the underlying is 10%, then the price of the underlying during the year will move between

- -10% and +10% with 68% probability,
- -2 x 10% and +2 x 10% with 95.4% probability,
- -3 x 10% and +3 x 10% with 99.7% probability.

The table below summarises the statistical volatility for different periods. It is interesting to see that the 10 day's 19.34% value is small. However, the 43.03% for the 120 days is relatively high. The chart illustrating volatility shows that the 30 day volatility has decreased from 72% to 19.83% from October to March. The decreasing volatility is accompanied by an increasing share price.

Historical volatility								
Period	10	20	30	60	90	120	150	180
Volatility	19.34	21.74	19.84	20.81	28.21	43.03	41.50	42.47



Historical volatility, IV index and market prices for the IBM shares

To conclude, historical volatility measures the relative price movements of the underlying. It is a *mathematical expression*, which shows (in a % value) how much the product's price has moved upwards or downwards in the examined period. The statistical volatility of a product is *constantly changing*. The value should be recalculated on a daily basis, because the results can change significantly from one day to the next. Thus, it is common to calculate a 10 or 20 day moving average to help future analyses. This moving average will be used later to calculate *fair value* for the different option valuation models. In practice, it can be used to estimate how probable is a given change in market prices. If the statistical volatility and the current market price of a future contract are known, then the probability of the price reaching a certain level can be calculated.

4.10. Implied Volatility

Implied volatility is the presumable size of the market price movements, when the option premium is known. In an option valuation model when *K* indicates the strike price, *S* indicates the current stock price, *T* indicates the time until option expiration, and *r* indicates the risk-free interest rate, the implied volatility is calculated when the theoretical equals the actual market premium. *An option's implied volatility is the volatility that makes the theoretical premium and the market premium equal*. The value of *S* stock price, *K* strike price, and *T* time is trivial. The size of the risk-free interest rate for *T* duration is also obvious. Volatility is more complicated. When estimated from historical data, the results are largely influenced by the time horizon. Volatility can be considered constant only in short term. In long term, no option's volatility can be said to be constant. When an option is traded on the market, then from the known premium one can calculate what value of volatility would make the theoretical and the market premium equal with the help of the Black-Scholes model. This is called statistical or historical volatility.

Let's see an example. An investor would like to calculate the implied volatility of a 875 Call strike price. The premium is 6.75 cents. The underlying costs 812.5 cents and the time until expiry date is 36 days. A treasury bond's return (with similar maturity to the option) is 7%. First the investor uses the following 4 volatility values in the Black-Scholes model: 20, 25, 30 and 35%. Then he gets the theoretical premium value. He will see that the volatility of the 875 Call option is around 25%. The market expects the price to change that much. Market analysts execute these calculations for every option in a given class, then weight the strike prices with the size of the volume. This way, the implied volatility for the whole option class can be measured. There is no universal formula to calculate implied volatility. The Black-Scholes model is not a perfect solution to solve this problem. The implied volatility can be calculated as shown in the example or with the assistance of sophisticated software.

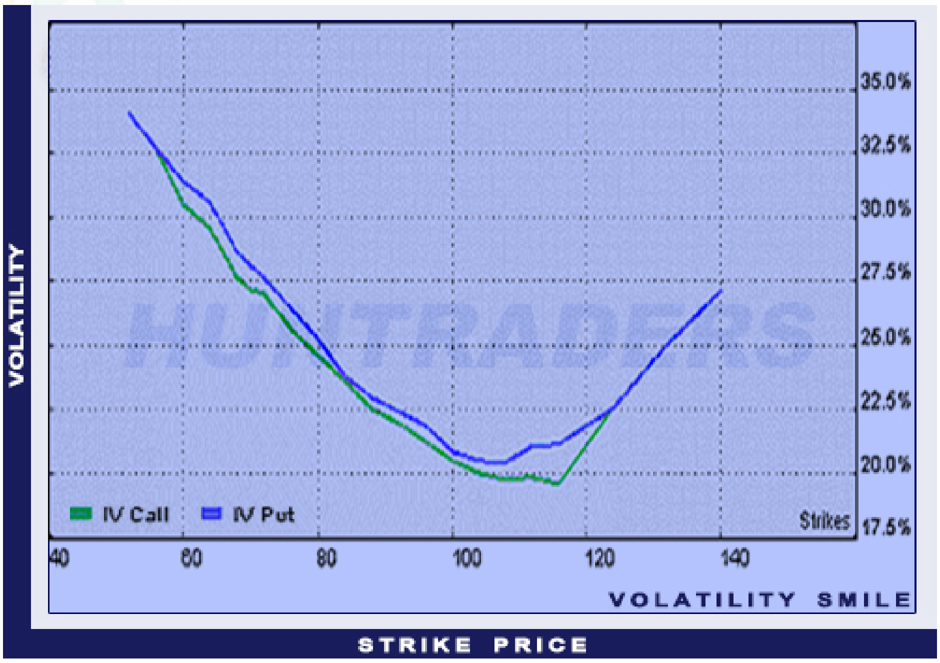
Implied volatility shows the option market participants' expected volatility for the next *T* time period. In the contrary, historical volatility shows volatility for a past period, thus it must be used carefully to assess future volatility. It is difficult to calculate volatility manually, but with Microsoft Excel the calculation can be solved within seconds. At first it may be hard to decide whether to focus on implied or historical volatility. *Usually the implied volatility* is calculated, because it is based on the consensus of market participants. Obviously, they can be wrong. When that happens, opportunities for free profit can emerge. Implied volatility usually affects the daily profit/loss of investors, so it is recommended to pay attention to the changes. Experienced traders usually follow the changed in historical volatility as well, especially when its value differs greatly from the implied volatility. Many think it is a sign. *When the implied volatility is above historical volatility, they sell options, and buy option when the implied volatility is below the statistical*. The divergence between the two volatilities should be assessed every time, because it is not always clear which one has the higher and which one has the lower value. There is a solid reason when the market pays more for implied volatility than the size of the historical volatility would justify.

To conclude, implied volatility is the *key* to earn profits regularly on the market. Implied volatility is a *calculated value*, which concerns only the option, not the underlying. It is the *expected volatility in the future*, one may say it is the *buyer/seller demand for a given option*. To use it in practice, it is crucial to calculate the *rate of the volatility*. Usually one of the following is used: 20-day-long volatility proportion (value of 1-day implied volatility divided by the value of the 20-day historical volatility) and 90-day-long volatility proportion (1-day implied volatility / 90-day historical volatility). When implied volatility is higher than historical volatility, then the rate is high and the *option is overpriced*. When implied volatility returns to the normal level, the price of the option will decrease as well. This is why investors buy Put options when they expect the underlying's price to decrease. When the option has a high implied volatility (it is overpriced), and the price of the underlying decreases, the value of the Put option will also decrease. It is a similar situation when one expects the underlying's price to increase and buys a Call option. When everyone expects the prices to grow, the market participants can easily make the options overpriced. To avoid this, one should first find an underlying with the help of the chart analysis which is expected to have price changes. Then the direction of the expected price change should be determined. After that it should be evaluated how the market has priced the option (fair, under- or overpriced).*Avoid buying overpriced and selling under-priced options, as it will only result losses*. One should try selling overpriced and buying under-priced options. *Time decay* will be negligible when buying under-priced options which are farther from the expiry date. When an underlying's price start changing, the demand for options will grow along with the implied volatility and it will result an increase in the option price. When selling options, it should be overpriced and close to the expiry day. It makes the decay larger and the investor can receive the option premium.

4.11. Volatility smile

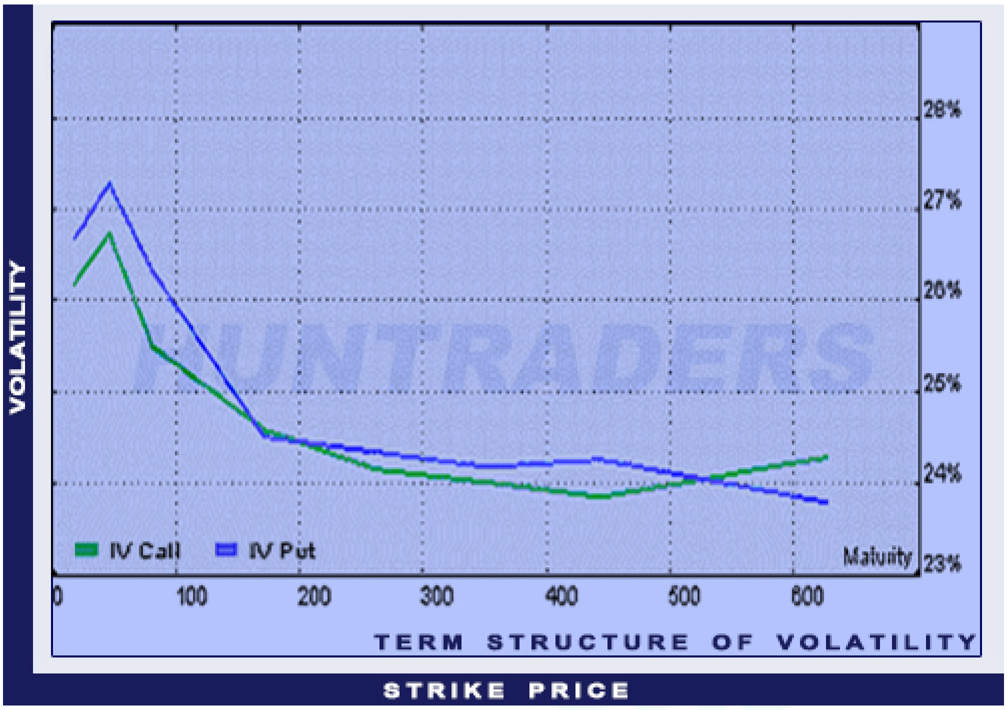
One might think that an underlying has a specific volatility for a given period and the volatility of options with different strike prices equals to that value. It is easy to understand that the volatility is different for options with different expirations. However, it seems strange that options with same expirations but different strike prices have different volatilities. Yet, this is the case. This phenomenon is called the *volatility smile*. It is also important to introduce the *term structure of volatility*.

The **volatility smile** is the chart illustrating the volatility of options with the same underlying and expiration and different strike prices. The X axis shows the strike prices.



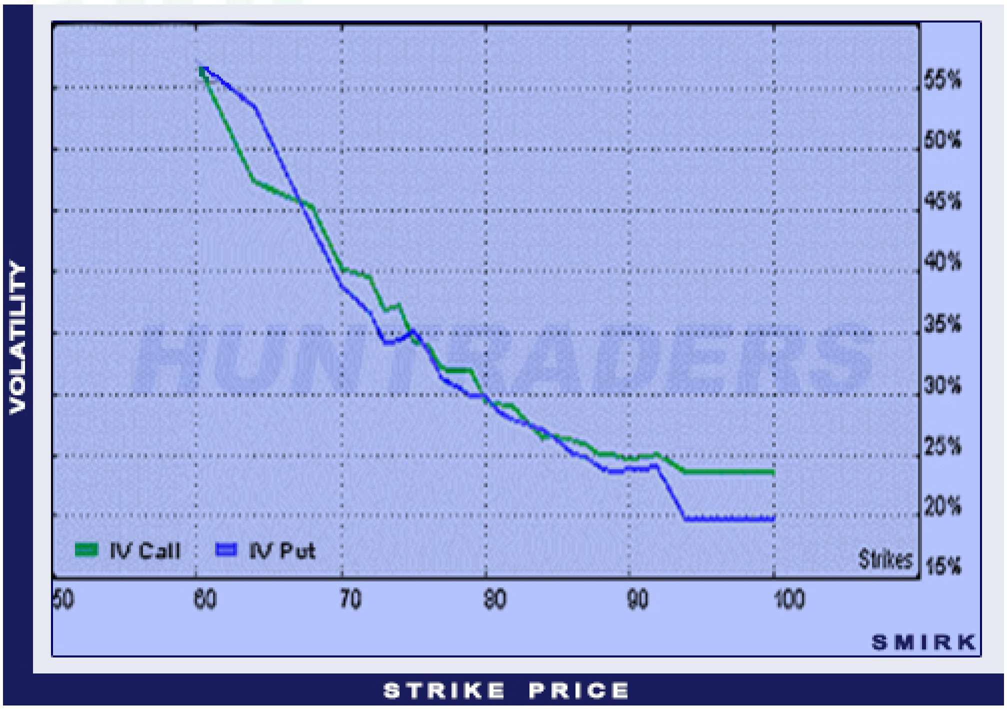
Volatility smile

The **term structure of volatility** is a chart illustrating the volatility of options with the same underlying and strike prices but with different expiry dates.



The term structure of volatility for ATM options on the DJX index

In the Black-Scholes model both charts are horizontal lines. In reality, the volatility is the smallest where the strike price equals the underlying's price. The volatility is higher where the strike is lower and higher than that point. In that case, the chart "draws" a smile. However, it can happen that one of the ends of the curve turns to concave. Then the smile is called a **volatility smirk**.



Volatility smirk on the DJX index (maturity: April 2003)

The **volatility matrix** is a table illustrating the volatility of options with the same underlying with maturity and strike prices being variables.

5. Bonds

5.1. Bond issuer

The bond issuer is the entity which is responsible to perform on the obligation defined by the security. Governments, corporations, and even central banks are able to issue bonds.

Government securities

Government bonds are issued by the Government. When buying a governmental bond, one lends money to the government with a prespecified interest rate and duration.

Investing in government bonds are considered to be safe, as the issuer is the state itself – one of the biggest and most stable player on the financial market. At maturity, the state guarantees to pay back the principal amount along with the outstanding interest rates.

There are several benefits of investing in government bonds:

They are *safe*, because the payments – including both the principal and the interest rates – are guaranteed by the government. These claims never expire; they have no forfeiture deadline.

Governmental bonds are sold on market price before maturity. Bonds are *flexible* investments, as there are multiple possibilities (regarding duration and face value) in circulation on the market.

Corporate bonds

With buying corporate bonds (or corporates) one can lend money to firms with a prespecified duration, in exchange of periodically paid interests. Duration means the period of time in which the corporation pays interest rates periodically. The investor cannot withdraw the principal until the end of that period. The firm undertakes to repay the principal (also called face value) at maturity.

Compared to government or treasury bonds, corporates have higher returns, but are riskier.

Despite the risk, corporate bonds also have benefits:

One can easily *choose the desired degree of risk and return*. There are firms officially considered safe by credit rating agencies. Corporates usually pay higher returns than government bonds in exchange for the moderate risk and capital guarantee. The difference between the two types of bonds can reach up to 2-3% on a yearly basis.

Corporate bonds can be sold anytime, if one needs the invested capital before maturity. When withdrawing a bank deposit, all interest payments are going to be lost. However, selling corporate bonds before maturity may even result extra profits.

5.2. Duration

Duration of a bond is the period for which the issuer undertakes the obligation to pay interests (coupons) and repay the principal (face value) at the end (at maturity).

The longer the duration, the more sensitive the market price of the bond to changes in the interest rates on the market. Hence, investors generally expect higher returns for bonds with longer duration. When market participants expect an increase in the interest rates over time, the interest rates are going to be higher for bonds with longer duration than for the short-term ones. Furthermore, the farther the maturity date the more inflation indicators and credit risk analyses influence the market prices. That is why investors (in case of long-term investments) prefer bonds with larger coupons and higher returns.

Short-term investments can be easily withdrawn and reinvested in bonds with more beneficial interests. However, investors interested in long-term deals are tied to the initial interests. The additional compensation on the return is called *"liquidity premium"* and it is the reason why long-term investments come with higher returns than similar short-term investments.

At maturity, the last cash flow is due, which means the end of the bond's "life".

In case of perpetuities, the security has no maturity; its duration is infinite. In turn, the face value will never be repaid and the investor's only source of cash flow is exclusively made up by the perpetual interest payments.

Short-term bonds

Short-term bonds' maturity is *no more than 1 year*. There may be bonds with longer maturity than a year, but then shorten their duration to less than a year. These bonds belong to this group.

Medium-term bonds

In this category, the maturity is typically *between 1 and 5 years*.

Long-term bonds

Long-term bonds are issued with the maturity of *at least 5 years*.

Perpetuities

Perpetuities have no maturity.

5.3. Price of bonds

Bonds are issued with various face values (also called nominal value, par value, and principal). By examining the market price, it may not be obvious whether the bond costs currently less or more than its nominal value. The market price is usually expressed as a percentage of the face value, so the market price does not need to be indicated separately.

Premium Bonds

Premium bonds' market price is above their face value. Then the price will depreciate over time.

Discount Bonds

Discount bonds are issued for less than the nominal value and the price will eventually appreciate.

Example: if a bond's market price is EUR 102 and its nominal value is EUR 100, the bond trades for 102%.

Obviously, this calculation works backwards also. If the bond trades for less than 100% (market price < face value), one can assume that investors expect higher returns than the promised interest rate payment.

Central bank rates (central banks' interest rate) are constantly moving. Bonds neutralise these movements with the changes in market prices. When the interest rate increases, the price of the bond will decrease and the investors will have higher income. The price of the bond and the yield have an inverse relationship.

The closer the bond to its maturity the closer the price gets to the par value. However, unexpected circumstances around the issuer may affect the price of the bond.

The biggest advantage of premium bonds is the high interest rate. Premium bonds are less sensitive to the changes in the risk-free interest rate than discount bonds, and the investor receives more money during interest payments.

It is only possible to profit from higher market prices, if one can purchase the bond for a lower price.

5.4. Face value of bonds

Bonds are issued at the nominal value, which may be above or below the market price.

The nominal (or face) value of a bond is the "sales unit" price printed on the security. The bond can be purchased at this price when it's issued. The nominal value can differ from the market value.

When issuing a bond with *fixed coupon*, the issuer promises to pay the same amount of interests (expressed as a percentage of the nominal value) on prespecified dates. At maturity, the principal along with the last interest payment will be repaid.

Zero coupon bonds do not pay interests. These bonds are sold for less than the nominal value, and the issuer promises to repay the nominal value at maturity.

The cheaper the bond, the wider range of investors it can target, and the easier to find sellers or buyers.

Small investments

The nominal value of the bonds is between $1,000 and $10,000 (or other currency in the same range of value).

Average investments

The nominal value of the bonds is between $10,000 and $50,000 (or other currency in the same range of value).

Large investments

The nominal value of the bonds is above $50,000 (or other currency in the same range of value).

5.5. Frequency of interest payments

The interest rate of a bond is the nominal interest, projected to the nominal value and paid on prespecified future dates.

In case of *fixed interest* rate bonds, the interest rate does not change until maturity.

Buyers usually prefer bonds with higher frequency of interest payments, simply because payments are received more often.

In terms of frequency, US bonds generally pay interests semi-annually.

Quarterly payments

This frequency is only used rarely, only in special cases.

Semi-annual payments

Fixed coupon bonds usually pay interests semi-annually.

Annual payments

In case of annual coupon payments, the *interest for the whole year* is received yearly, on a prespecified date.

5.6. Country risk

Bonds can be distinguished on the basis of the country of origin (market and the headquarters of the issuer).

Bonds can be issued in any foreign country. They are called *international bonds*.

A foreign bond is issued on the financial market of a foreign country, and the price is determined in the currency of the given country.

The creditability of the foreign country is a key factor in this case. A country's creditability depends on economic fundamentals.

Developed countries

Countries with developed economy and legal system belong to this group, such as the countries of Western Europe. The GDP per capita and the value of human capital are the largest in these countries worldwide. The capital markets in these countries are regulated effectively.

Capital markets work smoothly in Commonwealth countries. People who wish to invest are able to give their savings directly to firms. On the European market, the intermediation of banks is significant. People place their savings in banks and companies request financing from banks.

The market of bonds in Hungary is not fully developed yet. Apart from bonds issued by banks, MOL, E-STAR and Business Telecom bonds are in circulation on the market. In Hungary, short-term government bonds are called treasury bills. They usually have duration between 3 and 12 months.

The bond market is the most developed in the *United States*. The US has such a stable bond market, that the average duration of bonds is 13-14 years, but there are even bonds with maturity of 30-50 years.

The corporate bond market in the *United Kingdom* is also developed, but bank loans are more important for business financing.

The political and historical events between the two world wars affected the *German* corporate bond market to a large extent. After the introduction of the Euro, more and more firms have attempted to receive funds from the capital market as well.

Less developed, developing or emerging countries

The biggest economies in this group is the so-called BRIC countries. Eastern European countries, the most developed African nations, Latin-American states and Middle East belong to this group as well.

MINT countries (Mexico, Indonesia, Nigeria, and Turkey) and Next Eleven countries (Bangladesh, Egypt, Indonesia, Iran, Mexico, Nigeria, Pakistan, Philippines, South Korea, Turkey, and Vietnam) are also included in this group.

Many investors may find the undeveloped banking system and financial markets unattractive. Companies can access funds with more difficulties, and that is why they cannot develop at maximum potential. Sometimes the lack of access to funds can even result the bankruptcy of those firms. The bankruptcy of banks is an additional risk.

The markets of these countries are usually smaller. Because of the smaller number of players, the liquidity on these markets is not adequate, especially in case of panic situations. Putsches and legal changes are more common thus market risk is not negligible.

5.7. Foreign exchange risk

Foreign exchange risk (also called FX or currency risk) applies to foreign investments. When there are foreign bonds in the investment portfolio, the return is not only dependent on the fluctuations of market prices on the foreign markets, but also on the fluctuations of the exchange rates between the domestic and the foreign currencies.

The weakening (depreciation against the foreign currency) of the domestic currency is beneficial for the return of the foreign investment, because the currency exchange generates extra returns. When the domestic currency appreciates, the return of the investment decreases, because the performance of the investment worsens in terms of the domestic currency.

When investing in foreign governmental and corporate bonds, one must count with the enhanced FX risk exposure. Investing in American bonds or shares makes investors own securities with US dollar values. Investing in Indian shares makes investors own shares with Indian rupee values.

Hungary is embedded into the European Union's economy thus the exchange rate of Euro and Hungarian Forint is not as volatile as Forint against other exchange rates.

Some firms in the public and private sectors may decide to issue bonds or other securities in foreign currency. It can happen for plenty of reasons. The most important three are the following: decreasing FX risk, increasing funds only available abroad, and increasing trust in the given bond (and in the currency and solvency as well). There are 13 important foreign bonds:

- Eurodollar bond: a US dollar-denominated bond (issuer: non-US entity outside the US market)
- Kangaroo bond: an Australian dollar-denominated bond
- Maple bond: a Canadian dollar-denominated bond
- Samurai bond: a Japanese yen-denominated bond (issuer: non-Japanese entity on the Japanese market)
- Shibosai Bond: a private placement bond in Japanese market with distribution limited to institutions and banks
- Yankee bond: a US dollar-denominated bond (issuer: non-US entity on the US market)
- Shogun bond: non-yen-denominated bond (issuer: non-Japanese institution or government on the Japanese market)
- Bulldog bond: a British pound-denominated bond
- Matrioshka bond: a Russian rouble-denominated bond
- Arirang bond: a Korean won-denominated bond
- Kimchi bond: a non-Korean won-denominated bond
- Formosa bond: a non-New Taiwan Dollar-denominated bond
- Panda bond: a Chinese renminbi-denominated bond

5.8. Sector (industry) related risk

Sector is an industry or market, in which the participating companies have similar operational traits.

Investors tend to invest in companies of specific industries in order to achieve their financial goals. Despite the instability, investing in companies of other industries may result larger returns. Investing in several sector concerns investors with larger risk appetite.

Services sector

The services sector deals with the “production” of immaterial goods. Instead of goods, these firms provide services. It has numerous branches, such as warehousing, transporting, information services, investment services, professional-, technical-, scientific services, waste management services, healthcare and social services, arts, entertainment, and leisure services.

The social development and the wealth accumulation of citizens have enabled developed countries to extend their services network and provide high quality services.

The USA, the United Kingdom, Australia, and China emphasise their services sectors. In the US, the ISM monthly index indicates the business activities in the services sector.

Services are currently the leading sector: 61% of the world’s GDP is produced in this sector.

Walt Disney is a good example of the services sector.

Industrial Goods sector

Industrial goods are used in the construction and manufacturing sectors. Businesses in the manufacturing sector mostly deal with aircraft production, military equipment production, industrial machines and tools production, and construction. Cement and steel production belong to this sector as well. The performance of the manufacturing sector largely depends on the demand and supply of the constructions on the real estate market.

The sector’s cyclicality can be divided into two parts: accelerated or slowing growth and shrinkage. The best investment opportunities are in the accelerated growth and in the slowing shrinkage phases.

Governmental expenditures are especially important in this sector, as plenty of funds flow to businesses this way. Governments are big customers in the aerospace and military industries, and of the road, railway, and transportation infrastructure. The military and infrastructural investments are especially important within the sector.

Many of the biggest firms in the US all operate in the manufacturing sector. Examples include General Electric (GE), Boeing, and Caterpillar.

Consumer goods sector

Consumer goods sector includes firms providing fast moving consumer goods (FMCG), packaged goods, clothing, beverages, vehicles and electronics.

The performance of the sector depends on the consumers' behaviour.

The biggest American firms of this sector include Apple, Coca Cola, and Nike.

Basic Materials sector

The decisions of the clients of the firms in this sector are determined primarily by prices.

The income of the participating firms is defined by the commodity prices of basic materials, driven by supply and demand.

In the basic materials sector, a firm can gain competitive advantage when it can produce with low operating costs relative to other firms. The storage of materials, their access, and efficient production methods are the most common keys to success. Only a few firms can create these advantages, thus there are relatively few businesses competing in this sector.

This market is excessively cyclical. In times of flourishment, enormous amounts of funds flow into the sector, while in times of recession, many firms may shut down. A recession can affect firms with high operating costs more than the ones with competitive advantages.

The biggest American firms in the sector are Exxon Mobil, BHP Billiton and Alcoa.

Energy sector

In the past 15 years, renewable energy became the most dynamically growing segment of the energy sector. The increasing concern for climate change, the striving for energy security, the peak of the oil yield, the emergence of new technologies, and the appearance of environmentally conscious consumers are all factors responsible for the increase of the sector.

The price of raw materials influences the revenues in the sector, thus the sector is characterised by cyclicality and the profitability of the participating firms can be unstable.

Financial sector

Financial instruments currently not in use and accessible either in short or long term, are collected on the financial markets.

The financial sector is the place of exchanging money and the collection of players (individuals and institutes), financial instruments, mechanisms, regulations, regulators, and customs.

The sector's primary task is to transmit available financial instruments (or savings) to the users of the funds.

Therefore, on the financial market, funds change hands in different times.

The main institutions of the financial sector are the following:

Commercial banks: financial institutes with the broadest profile, they deal with almost all kinds of banking activities.

Depository institutes: they transform savings into long-term credit resources (e.g. savings banks financing housing constructions).

Insurances, pension funds: their main function is to pay compensations from the collected funds, or to make pension payments to the customers. To ensure the high-quality service, they need to invest the collected funds securely and profitably. The revenues, expenses, and the available financial instruments of pension funds are easy to quantify even in long term. Thus, they have the opportunity to invest in long-term with high yields, primarily in corporate bonds and shares.

Investment firms and investment funds: they specialise in collecting small funds from many small individuals intending to purchase shares. They invest the collected savings profitably with expertise and low transaction costs.

Financing firms: their revenues are from issuing shares and from loans. The capital acquired this way is then used to provide small, short-term consumer loans or business loans.

Other intermediary financial institutes: state agencies and institutions, brokerages, mortgage banks, and leasing companies.

Companies in the financial sector are heavily influenced by economic processes. They usually acquire capital by issuing new shares, weakening the existing shareholders' ownership in the firm.

The biggest financial businesses are for example the Citigroup, the MasterCard or the JPMorgan.

Healthcare sector

Healthcare is one of the few sectors that are related directly to human survival. New medical innovations can improve or extend patients' lives.

The sector has gone through an energetic growth phase in the past decade, especially in the United States. The pace of the growth has slowed down in the past few years. According to current trends, one can expect a decrease in the pace of growth of the sector. However, in the long term, the sector will remain powerful.

Healthcare includes all activities with the intension of

- preserving,
- restoring,
- stabilising one's health.

This sector is analysed, because the curing of people may rise many problems that can be solved by the different methodologies of economics.

Healthcare is a broad term, because apart from the healthcare system it also involves

- the production of pharmaceuticals,
- the pharmaceutical commerce,
- the manufacture and wholesale of medical aids,
- the financial and insurance supports related to incapacity of work,
- the biotechnology researches.

The biggest healthcare firms in the US are Johnson & Johnson, Pfizer and Sanofi.

Technology sector

IT is without doubt the sector that went through the biggest development in the past few decades.

Firms providing research and development (R&D) and technology-based services and products are the participants of the sector. Their activities include the hardware, the software, and the semiconductor industry.

The rapid development of technology makes it difficult to companies to gain sustainable competitive advantage in the sector. There is a huge potential in the industry, as the development is continuous.

The biggest American IT companies are Google, Cisco, and Microsoft.

Utilities sector

Utilities firms own developed infrastructures. Thus, they have large debts and are sensitive to interest rate changes. An increase in interest rate enlarges the capital expenses of the firms.

The biggest danger in the sector is the spread of renewable energy, which threatens the traditional utility usage levels.

Utilities firms are typically holdings, and they include the distribution, the production of electricity, and its infrastructure.

The biggest utilities companies are Duke Energy, American Electric Power and Westar Energy in the United States.

5.9. Bond credit rating

The biggest risk everyone must consider when investing in bonds is the credit risk. Credit risk is reflected in the price of the bond. Investors expect higher returns or other extra benefits from bonds with higher credit risk.

Credit risk can be decreased by buying senior bonds. Another way to minimize credit risk is to invest in diversified bonds.

Rating systems take into account both quality and quantity aspects.

Credit ratings can affect the yield of the bond, but also the issuer itself and the conditions of its future bonds.

There are two grades for bonds: investment grade and junk.

The grading of bonds is usually performed by three big credit rating agencies: Standard & Poors (S&P), Moody's and Fitch.

Investment grade

Investment grade concerns countries and borrowers which are worthy of investing.

The highest grade (for all 3 agencies) is *AAA* (prime). It indicates the most trustworthy, most stable, and the best borrowers with the lowest degree of credit risk.

The next category is *AA* (*AA+*, *AA*, *AA-*; high grade), which still indicates the reliability of bond issuers. The credit risk is low, but there is a bigger chance of default in long-term.

The next grade is the single *A* (*A+*, *A*, *A-*; upper medium) rating. Bond issuers with A rating are able to meet their financial obligations in most of the cases, but have a higher tendency to default.

The lowest grade in the investment grade category is *BBB* (*BBB+*, *BBB*, *BBB-*; lower medium). Firms and institutions belonging to this rating have the capacity to meet their obligations, but unfavourable economic circumstances may jeopardise the performance on those obligations.

Junks

Bonds rated as junks are usually not recommended to invest in, due to the high credit risk and unreliability.

Borrowers in the *BB* (*BB+*, *BB*, *BB-*; non-investment grade speculative) grade are economically more stable than the borrowers with lower grades. However, due to business and financial circumstances, they are characterised by continuous instability.

Entities with *B* (*B+*, *B*, *B-*; highly speculative) grade are more vulnerable in the long term, but currently can meet all their obligations without problem.

The *CCC* (*CCC+*; substantial risks, *CCC*; extremely speculative, *CCC-*, in default with little prospect for recovery) rating indicates entities currently economically weak. The fulfilment of obligations heavily depends on the current economic and financial state of their environment.

CC rating means current financial struggles and almost no prospects.

Entities in *C* category are financially distressed and at the edge of bankruptcy, but still able to meet obligations.

D is the worst credit rating, concerning insolvent borrowers, who cannot (or just partially can) meet their financial obligations.

The different credit rating agencies may rate borrowers differently. The agencies must be independent institutions; therefore differences in the ratings are normal.

5.10. Coupons

In bond trading, coupon is a word for interest rate. The coupon (nominal interest) is determined when the bond is issued and is expressed as a percentage of the face value.

The market price of a bond can differ from its initial price through its duration. The difference in prices is primarily dependent on the level of interest rates on the market.

The market's interest rates influence the price of the security inversely (in case of securities with fixed interest payments). When the interest rates decrease, the price of the security increases. This phenomenon only happens with fixed interest rates. The bond loses its attractiveness when the interests are increasing. In this case, bond owners are going to sell their securities.

Bonds with fixed coupons come with an extra risk, called interest rate risk. Interest rate risk is especially important when the interest rates on the market are strongly increasing. The changes in market prices are only important to the investor, if he or she does not intend to keep the bond until maturity. If the investor keeps the bond, then at maturity (if the issuer is solvent) the nominal value will be repaid.

Low coupons are between 0 and 2%, while **average coupons** promise returns between 2 and 4%. **High coupons** can have higher yield than 4%.

5.11. Bond interest rate

The interest rates are recorded in the contract. It is paid by the issuer after the nominal value of the security. Interest rates are usually fixed. The other case is when interest rates are floating, based on market conditions.

Fixed coupon bonds

Fixed coupon bonds have a fixed interest rate until the maturity (nominal interest). The interest payments occur usually annually. In some countries (e.g. in the US), the payments may happen semi-annually.

Floating coupon bonds

Floating rate notes (FRNs) pay variable interest rates instead of a periodic fix coupon. The issuer of the bond pays coupons after the interest periods (e.g. 3 months, 6 months, or 1 year). In the same time of the interest payment, the issuer announces the interest rate for the upcoming period as well. The interest rate is linked to reference rates, such as the EURIBOR (European Interbank Offered Rate) or the LIBOR (London Interbank Offered Rate).

The EURIBOR and the LIBOR are the rates, for which banks in the European Monetary Union and London are depositing money on short term. These reference rates show the amount of interests for which banks would lend money for short term. In terms of coupons, investing in FRNs are similar to other investments on the financial market, because the interest rates are periodically recorded, just like in the case of futures and forwards.